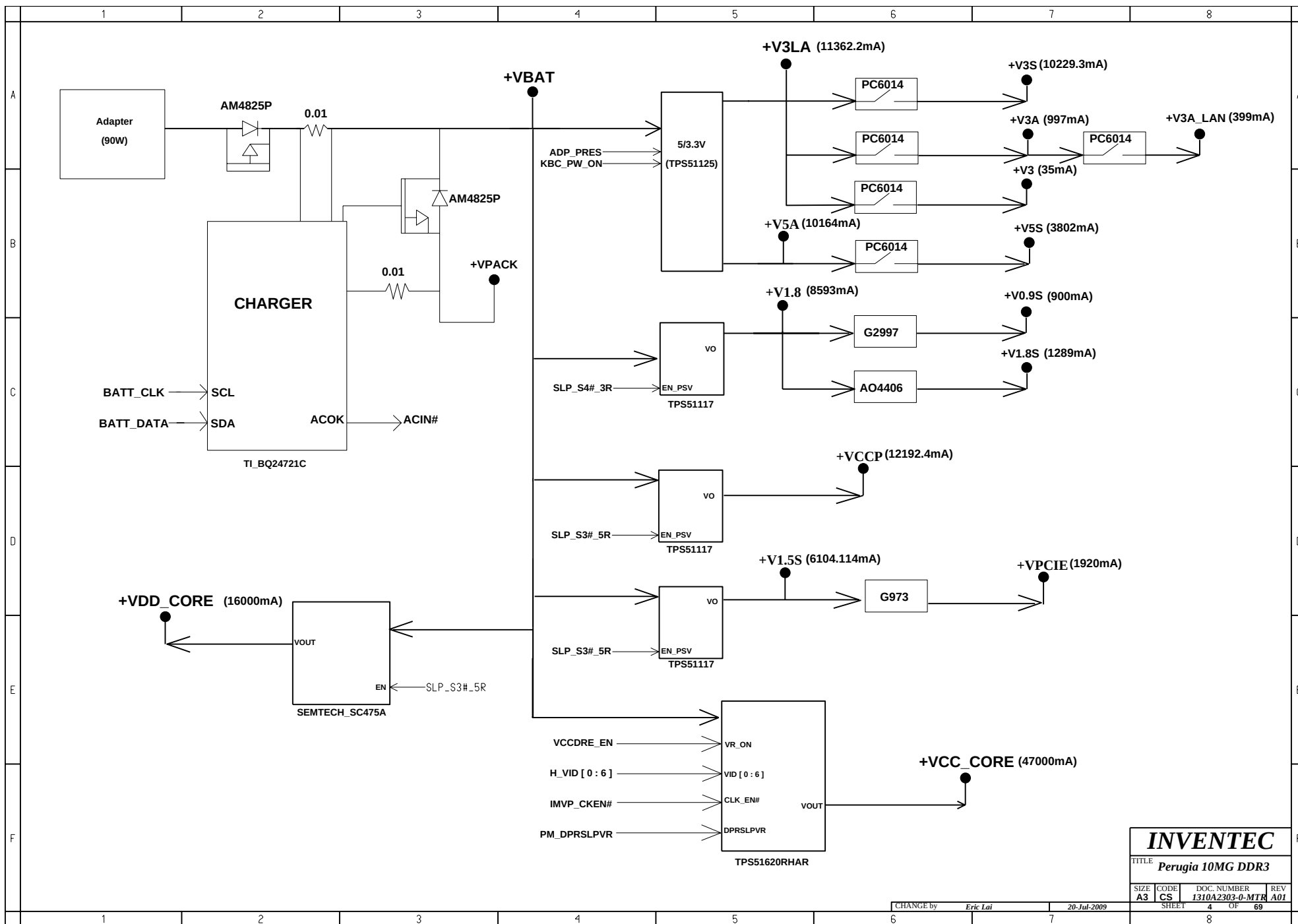


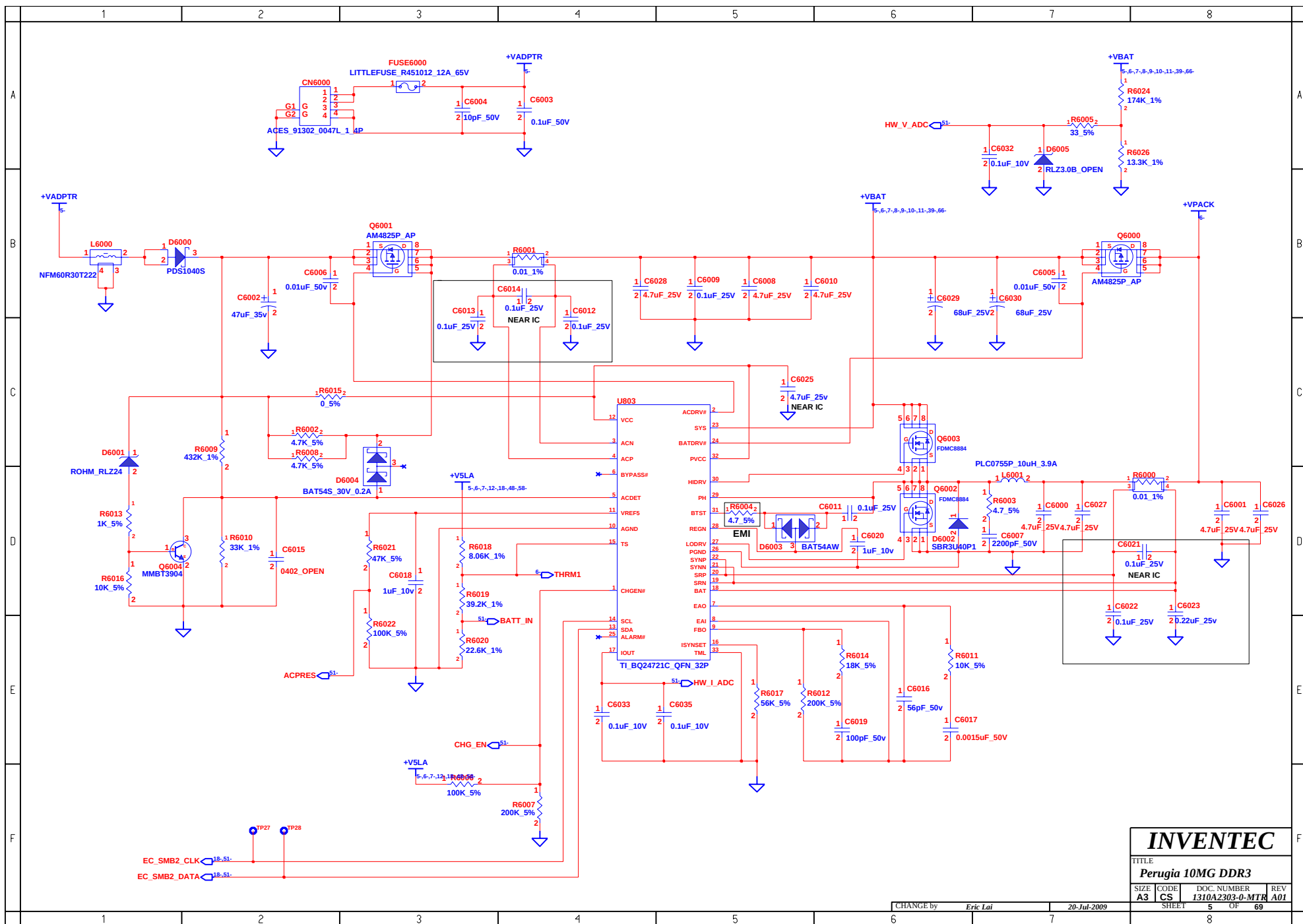
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A	THIS DRAWING AND SPECIFICATIONS, HEREIN, ARE THE PROPERTY OF INVENTEC CORPORATION AND SHALL NOT BE REPRODUCED, COPIED, OR USED IN WHOLE OR IN PART AS THE BASIS FOR THE MANUFACTURE OR SALE OF ITEMS WITHOUT WRITTEN PERMISSION, INVENTEC CORPORATION 2009 ALL RIGHT RESERVED																																																																																													
B																																																																																														
C	Perugia 10MG DDR3 CS Build(A01)																																																																																													
D	2009.07.20																																																																																													
E																																																																																														
F	<table><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td>DATE</td><td>CHANGE NO.</td><td>REV</td></tr></table>																									DATE	CHANGE NO.	REV			<table><tr><td></td><td>EE</td><td>DATE</td><td>POWER</td><td>DATE</td></tr><tr><td>DRAWER</td><td>Eric Lai</td><td></td><td>Bin-Zhi Lo</td><td></td></tr><tr><td>DESIGN</td><td>Eric Lai</td><td></td><td>Bin-Zhi Lo</td><td></td></tr><tr><td>CHECK</td><td>Eric Lai</td><td></td><td>Bin-Zhi Lo</td><td></td></tr><tr><td>RESPONSIBLE</td><td>Spring Liu</td><td></td><td>Bo-Yuan Shih</td><td></td></tr><tr><td colspan="5">SIZE = A3</td></tr><tr><td colspan="5">FILE NAME PG10MG-6050A2262201-MB-A02-09-20090519-DDR3</td></tr><tr><td>P/N</td><td colspan="4">6050A2262201</td></tr></table>		EE	DATE	POWER	DATE	DRAWER	Eric Lai		Bin-Zhi Lo		DESIGN	Eric Lai		Bin-Zhi Lo		CHECK	Eric Lai		Bin-Zhi Lo		RESPONSIBLE	Spring Liu		Bo-Yuan Shih		SIZE = A3					FILE NAME PG10MG-6050A2262201-MB-A02-09-20090519-DDR3					P/N	6050A2262201						<table><tr><td colspan="4">INVENTEC</td></tr><tr><td colspan="4">TITLE Perugia 10MG DDR3</td></tr><tr><td>SIZE</td><td>CODE</td><td>DOC. NUMBER</td><td>REV</td></tr><tr><td>A3</td><td>CS</td><td>1310A2303-0-MTR</td><td>A01</td></tr><tr><td colspan="2">SHEET</td><td>1</td><td>OF 69</td></tr></table>	INVENTEC				TITLE Perugia 10MG DDR3				SIZE	CODE	DOC. NUMBER	REV	A3	CS	1310A2303-0-MTR	A01	SHEET		1	OF 69
DATE	CHANGE NO.	REV																																																																																												
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SHEET		1	OF 69																																																																																											
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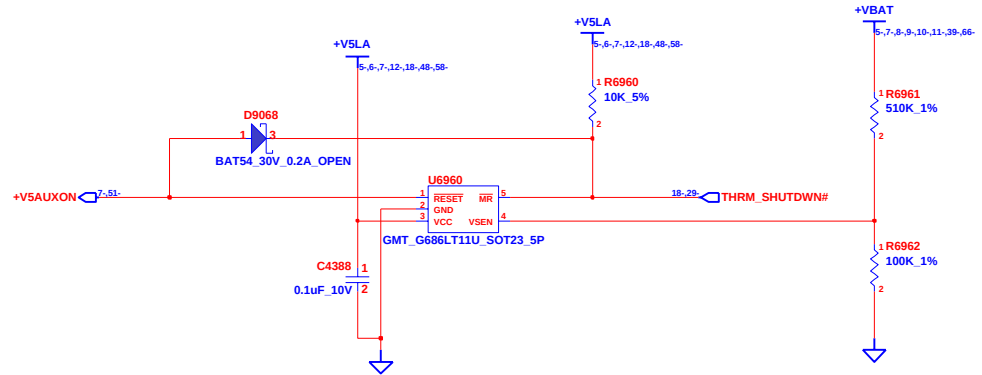
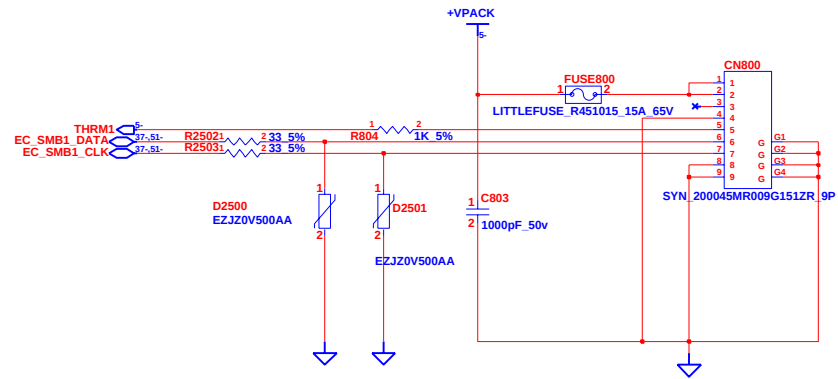
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- 55.LAN CONTROLLER
- 56.RJ45 & TRANSFORMER
- 57.HDMI & CEC
- 58.HDMI CONN
- 59.USB Card Reader
- 60.Express Card
- 61.WLAN /Debug
- 62.MDC 1.5 CONN
- 63.LED(M/B) & HOTKEY/B CONN
- 64.KILL SWITCH& HALL SENSOR
- 65.Screw
- 66.EMI
- 67.Hotkey board
- 68.Touch pad board
- 69.Power board



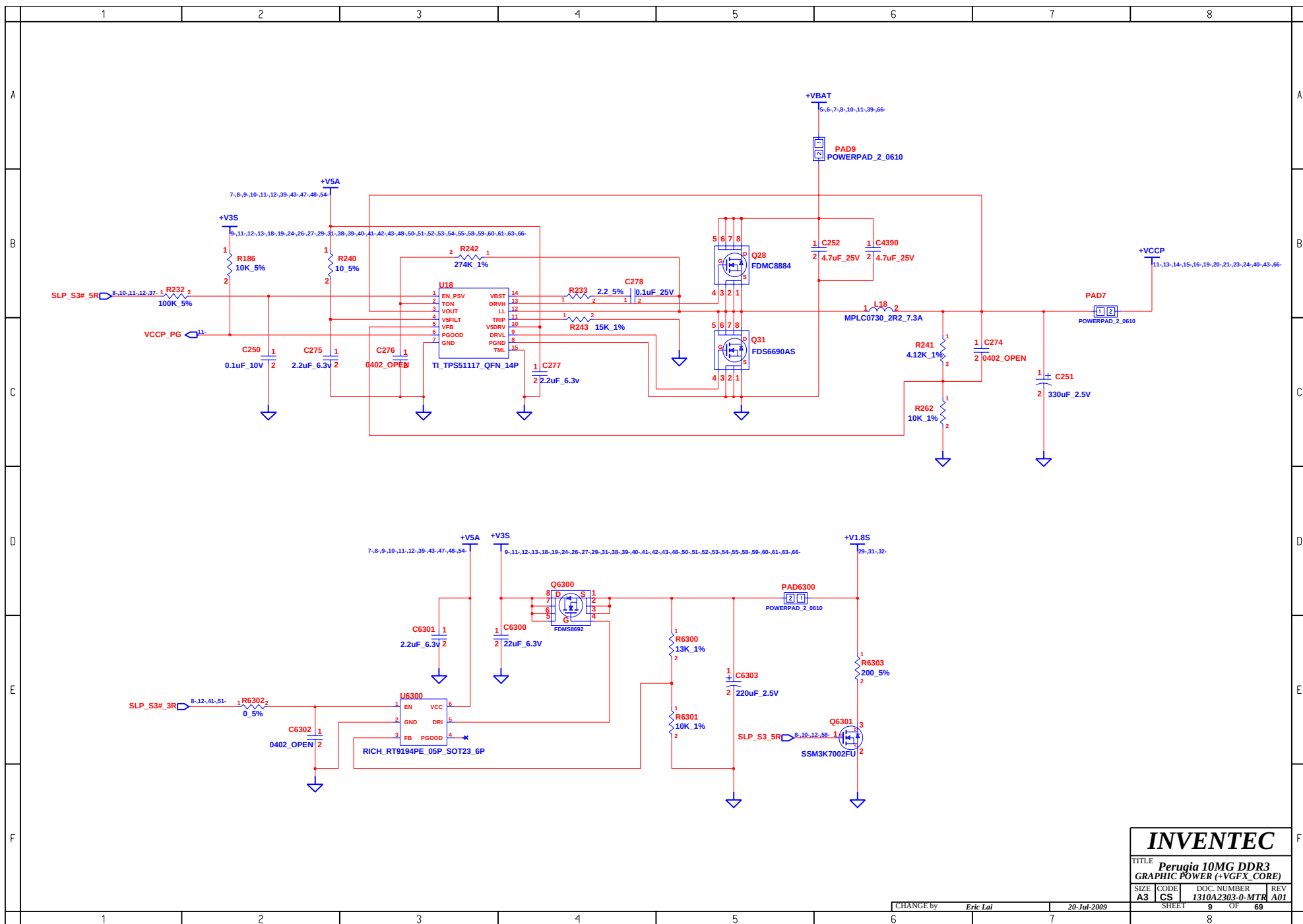
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TITLE Perugia 10MG DDR3			
SIZE A3	CODE CS	DOC. NUMBER I310AZ303-0-MTR	REV A01
CHANGE by Eric Lai		20-Jul-2009	
SHEET 4		OF 69	

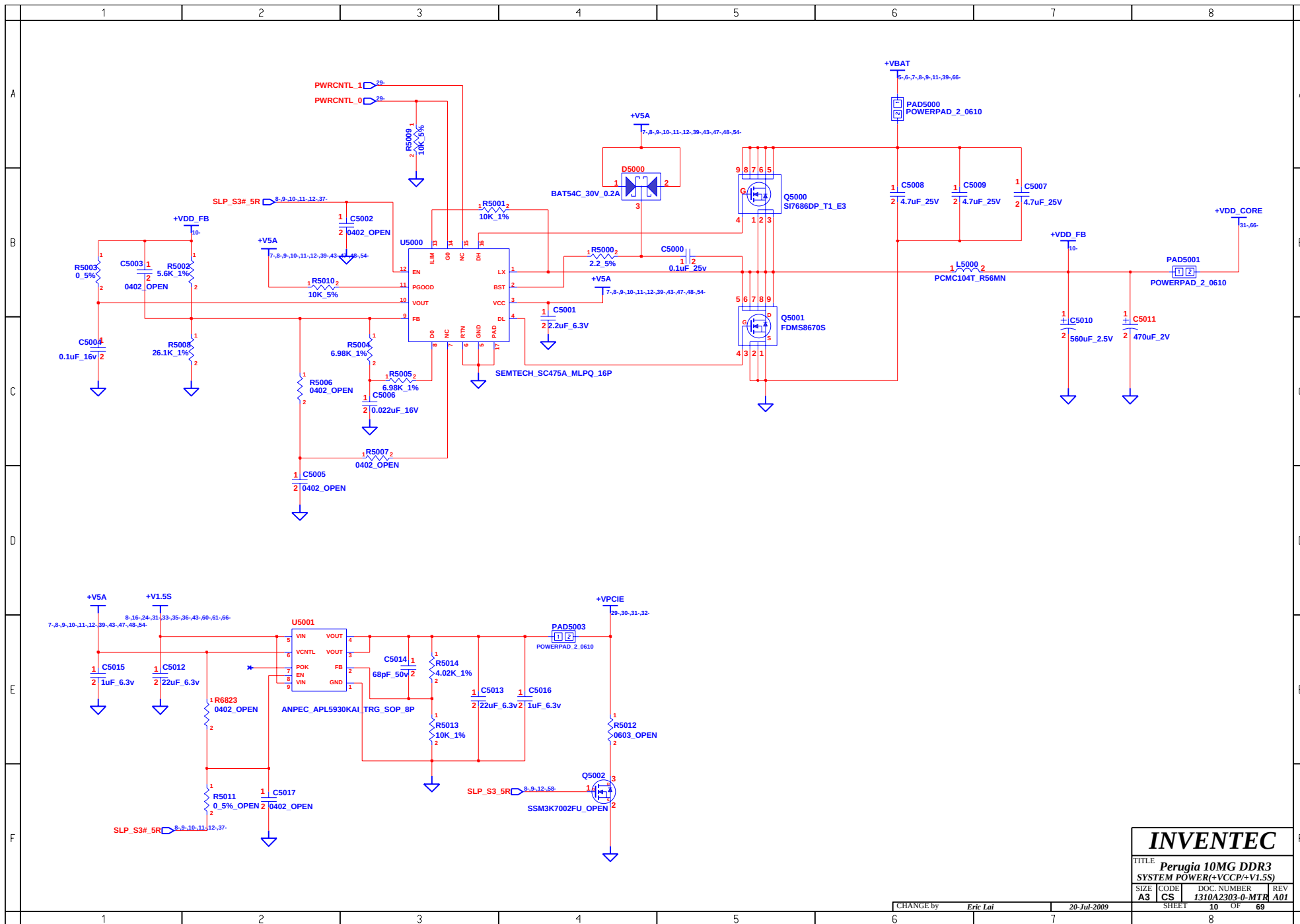


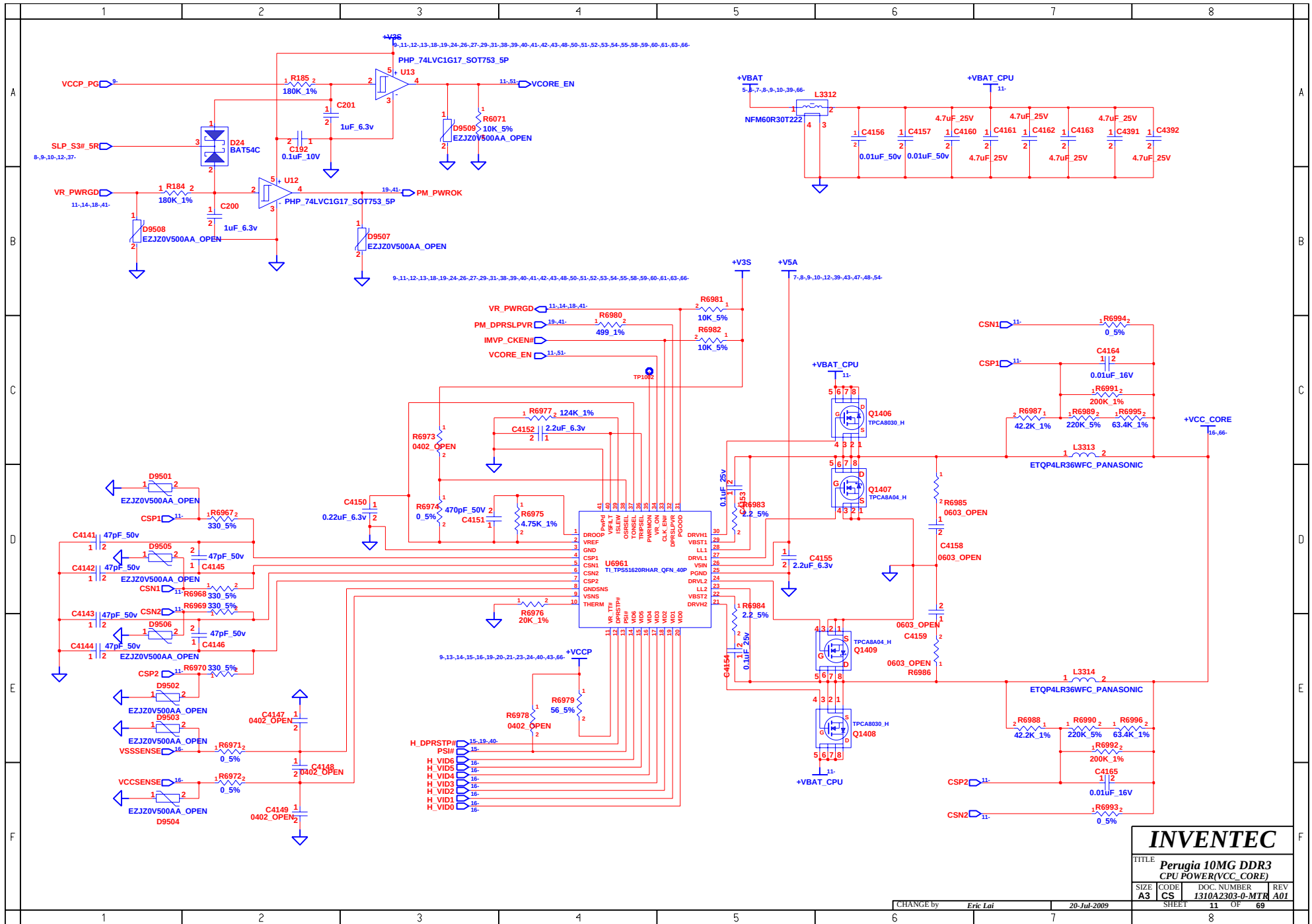


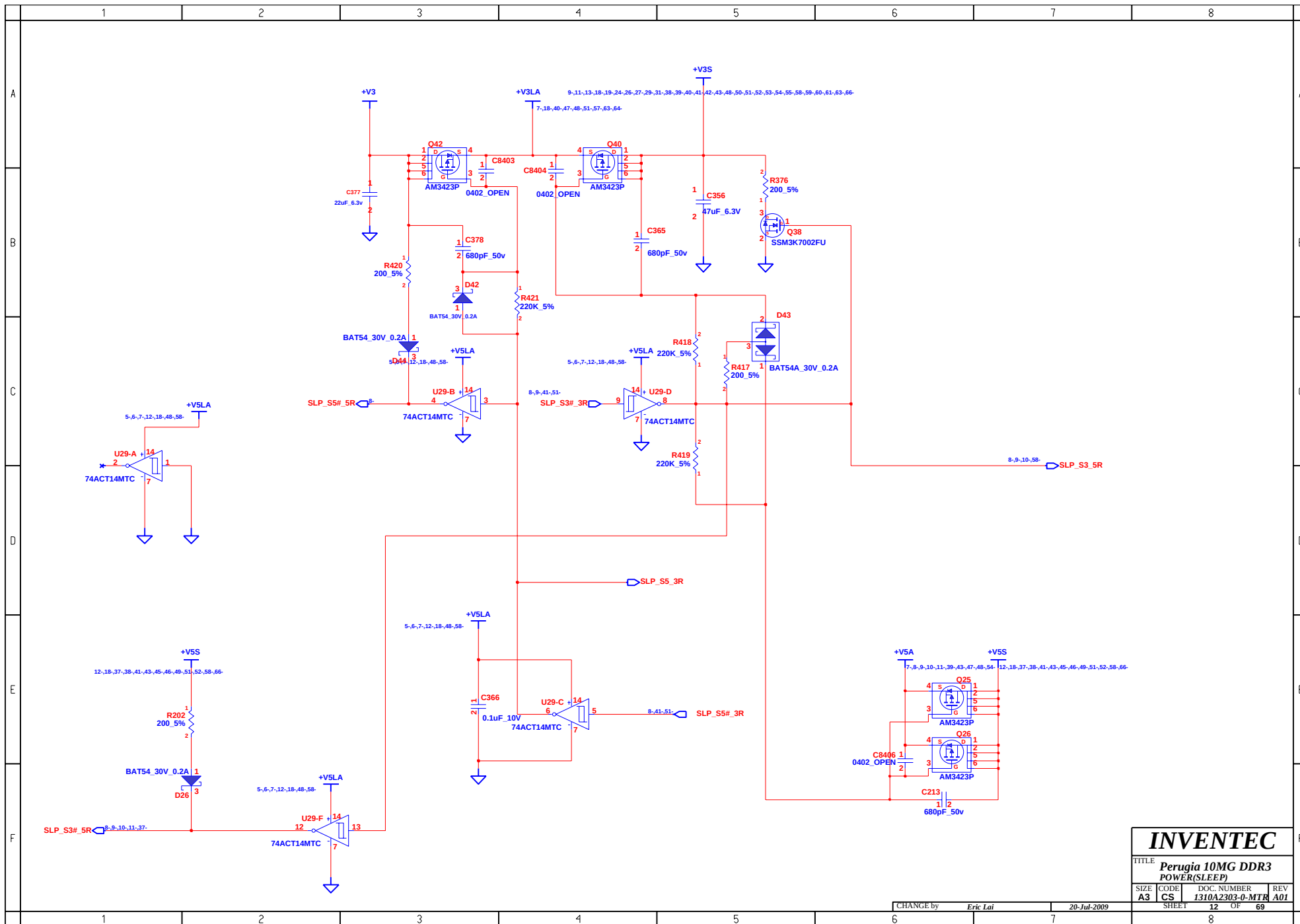
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TITLE			
SELECT & BATTERY CONN			
SIZE	CODE	DOC. NUMBER	REV
A3	CS	I310AZ303-0-MTR	A01
SHEET		6	OF 69

CHANGE by Eric Lai 20-Jul-2009









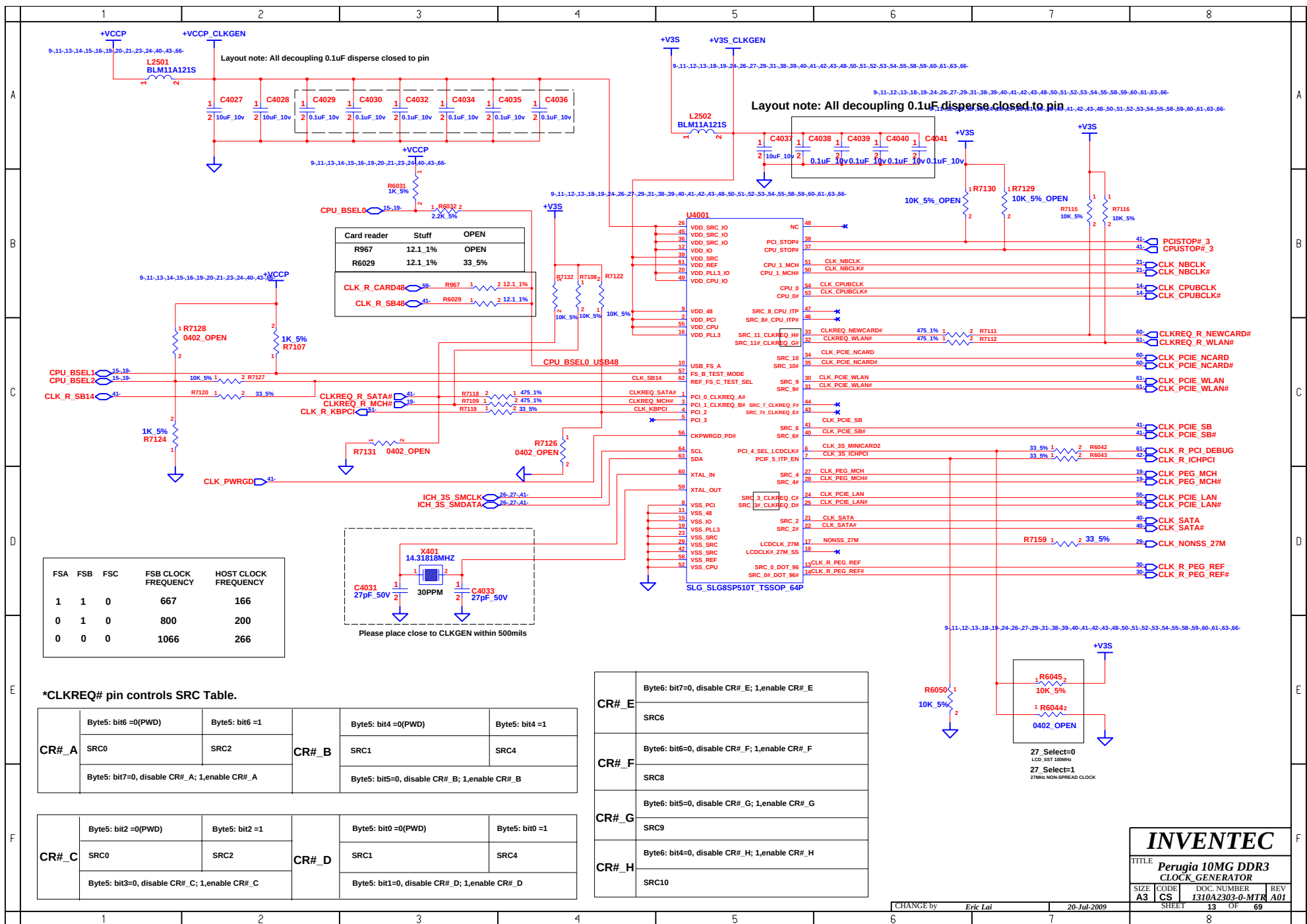
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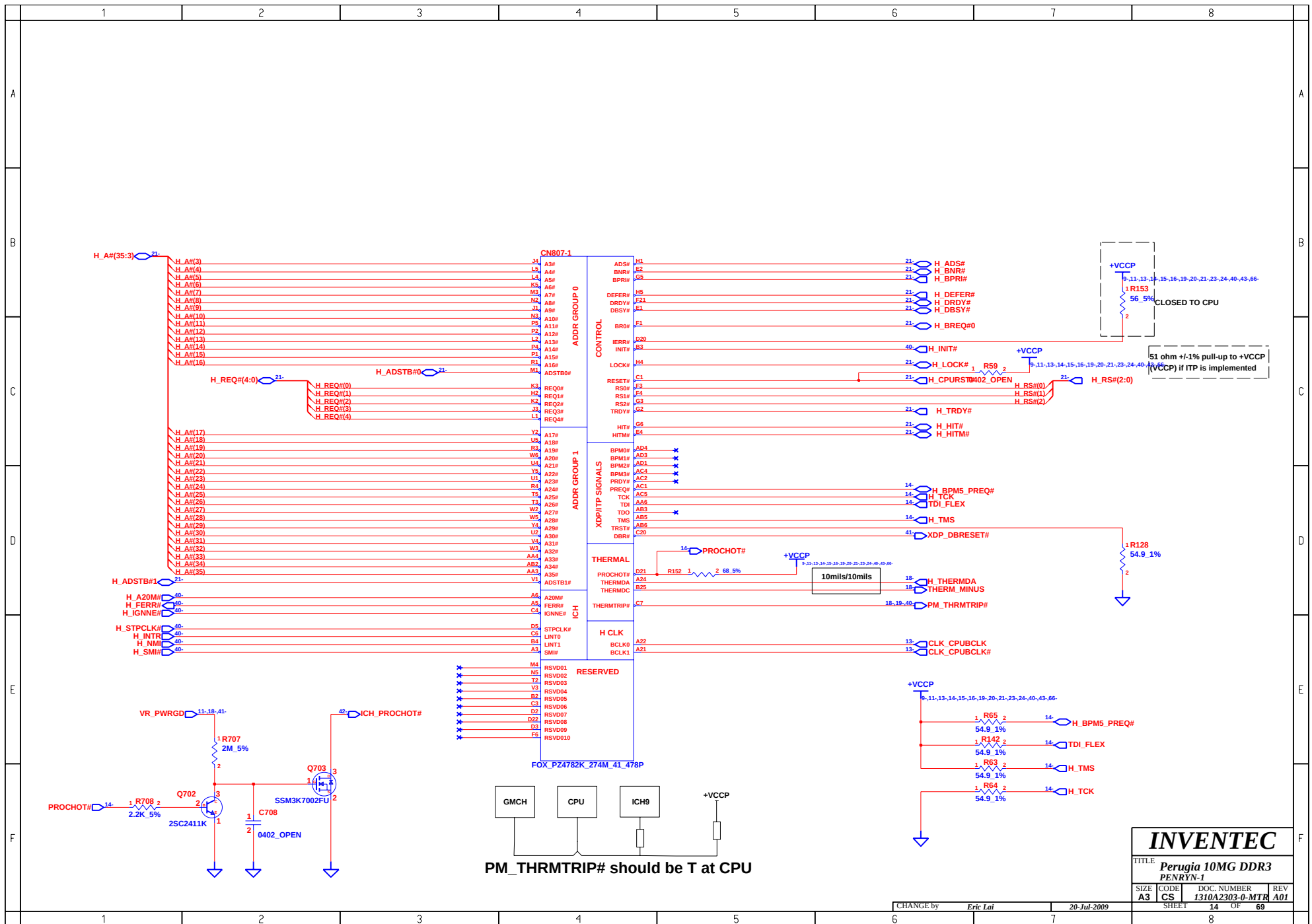
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**Perugia 10MG DDR3
POWER(SLEEP)**

SIZE	CODE	DOC. NUMBER	REV
A3	CS	I310AZ303-0-MTR	A01

CHANGE by Eric Lai 20-Jul-2009

SHEET 12 OF 69





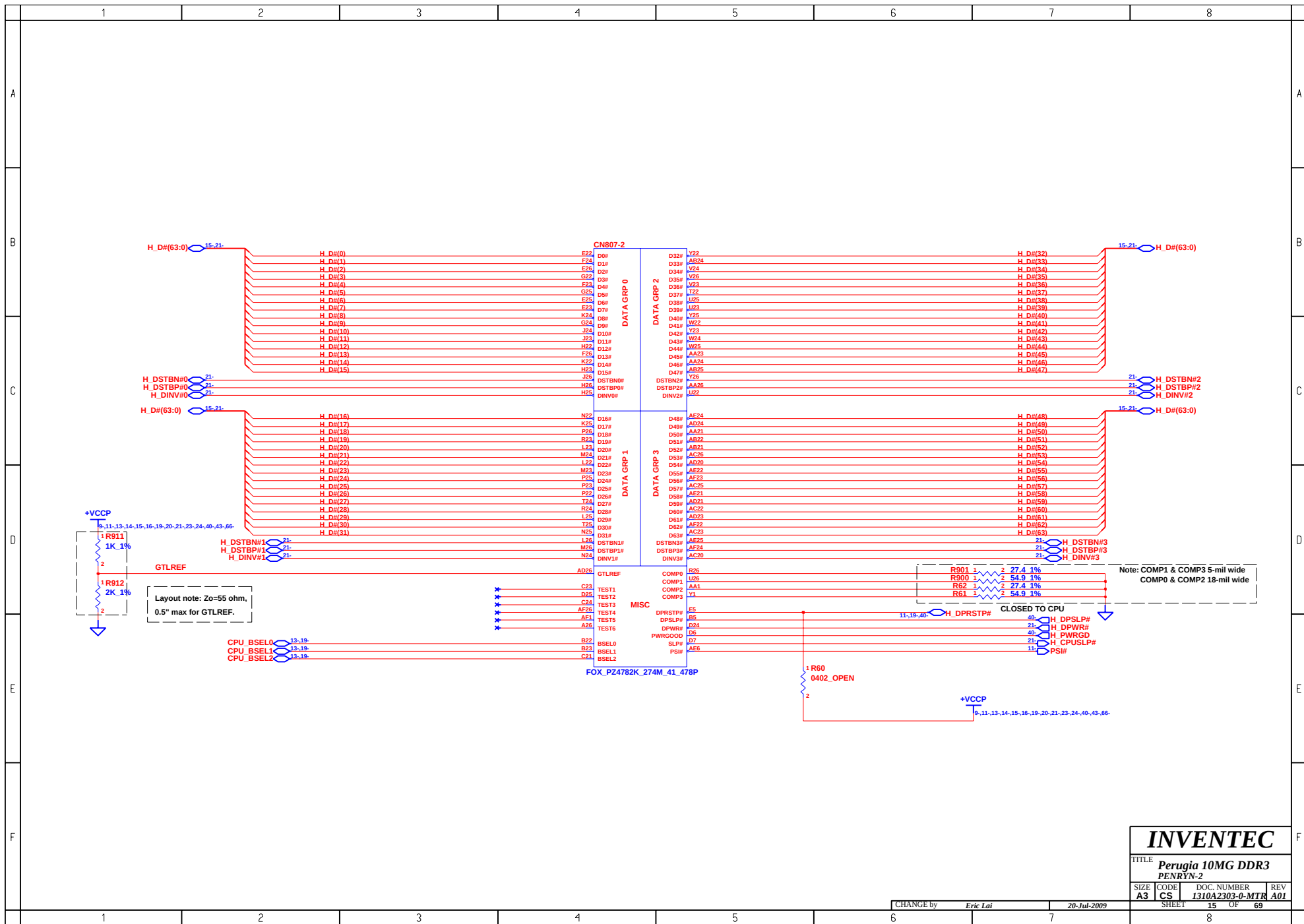
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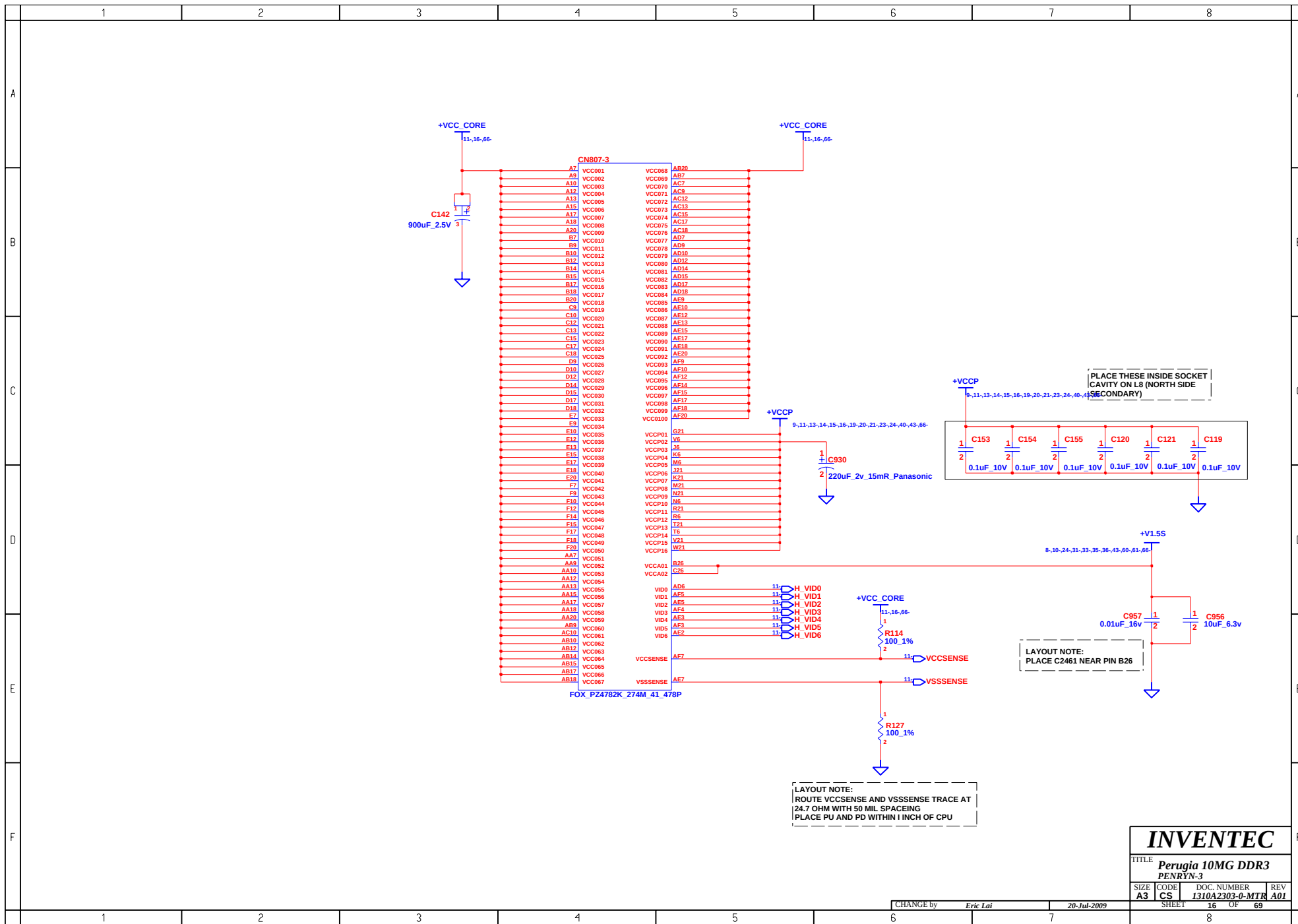
TITLE
Perugia 10MG DDR3
PENRYN-1

SIZE	CODE	DOC. NUMBER	REV
A3	CS	1310AZ303-0-MTR	A01

CHANGE by Eric Lai 20-Jul-2009

SHEET 14 OF 69





INVENTEC

TITLE
**Perugia 10MG DDR3
PENRYN-3**

SIZE	CODE	DOC. NUMBER	REV
A3	CS	I310AZ303-0-MTR	A01

CHANGE by Eric Lai 20-Jul-2009

SHEET 16 OF 69

	1	2	3	4	5	6	7	8	
A									A
B									B
C									C
D									D
E									E
F									F

CN807.4

A4

VSS001

A8

VSS002

A11

VSS003

A14

VSS004

A16

VSS005

A19

VSS006

A23

VSS007

AF2

VSS008

B6

VSS009

B8

VSS010

B11

VSS011

B13

VSS012

B16

VSS013

B19

VSS014

B21

VSS015

B24

VSS016

C5

VSS017

C8

VSS018

C11

VSS019

C14

VSS020

C16

VSS021

C19

VSS022

C2

VSS023

C22

VSS024

C25

VSS025

D1

VSS026

D4

VSS027

D8

VSS028

D11

VSS029

D13

VSS030

D16

VSS031

D19

VSS032

D23

VSS033

D26

VSS034

E3

VSS035

E6

VSS036

E8

VSS037

E11

VSS038

E14

VSS039

E16

VSS040

E19

VSS041

E21

VSS042

E24

VSS043

F5

VSS044

F8

VSS045

F11

VSS046

F13

VSS047

F16

VSS048

F19

VSS049

F2

VSS050

F22

VSS051

F25

VSS052

G4

VSS053

G1

VSS054

G23

VSS055

G26

VSS056

H3

VSS057

H6

VSS058

H21

VSS059

H24

VSS060

J4

VSS061

J5

VSS062

J22

VSS063

J25

VSS064

K1

VSS065

K4

VSS066

K23

VSS067

K26

VSS068

L3

VSS069

L6

VSS070

L21

VSS071

L24

VSS072

M2

VSS073

M5

VSS074

M22

VSS075

M25

VSS076

N1

VSS077

N4

VSS078

N23

VSS079

N26

VSS080

P1

VSS081

P6

VSS082

P21

VSS083

P24

VSS084

R2

VSS085

R5

VSS086

R22

VSS087

R25

VSS088

T1

VSS089

T4

VSS090

T23

VSS091

T26

VSS092

U3

VSS093

U6

VSS094

U21

VSS095

U24

VSS096

V2

VSS097

V5

VSS098

V22

VSS099

V25

VSS100

W1

VSS101

W4

VSS102

W23

VSS103

W26

VSS104

Y3

VSS105

Y6

VSS106

Y21

VSS107

Y24

VSS108

AA2

VSS109

AA5

VSS110

AA8

VSS111

AA11

VSS112

AA14

VSS113

AA16

VSS114

AA19

VSS115

AA22

VSS116

AA25

VSS117

AB1

VSS118

AB4

VSS119

AB8

VSS120

AB11

VSS121

AB13

VSS122

AB16

VSS123

AB18

VSS124

AB23

VSS125

AB26

VSS126

AC3

VSS127

AC6

VSS128

AC8

VSS129

AC11

VSS130

AC14

VSS131

AC16

VSS132

AC19

VSS133

AC21

VSS134

AC24

VSS135

AD2

VSS136

AD5

VSS137

AD8

VSS138

AD11

VSS139

AD13

VSS140

AD16

VSS141

AD19

VSS142

AD22

VSS143

AD25

VSS144

AE1

VSS145

AE4

VSS146

AE8

VSS147

AE11

VSS148

AE14

VSS149

AE16

VSS150

AE19

VSS151

AE23

VSS152

AE26

VSS153

A2

VSS154

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AF13

VSS158

AF16

VSS159

AF19

VSS160

AF21

VSS161

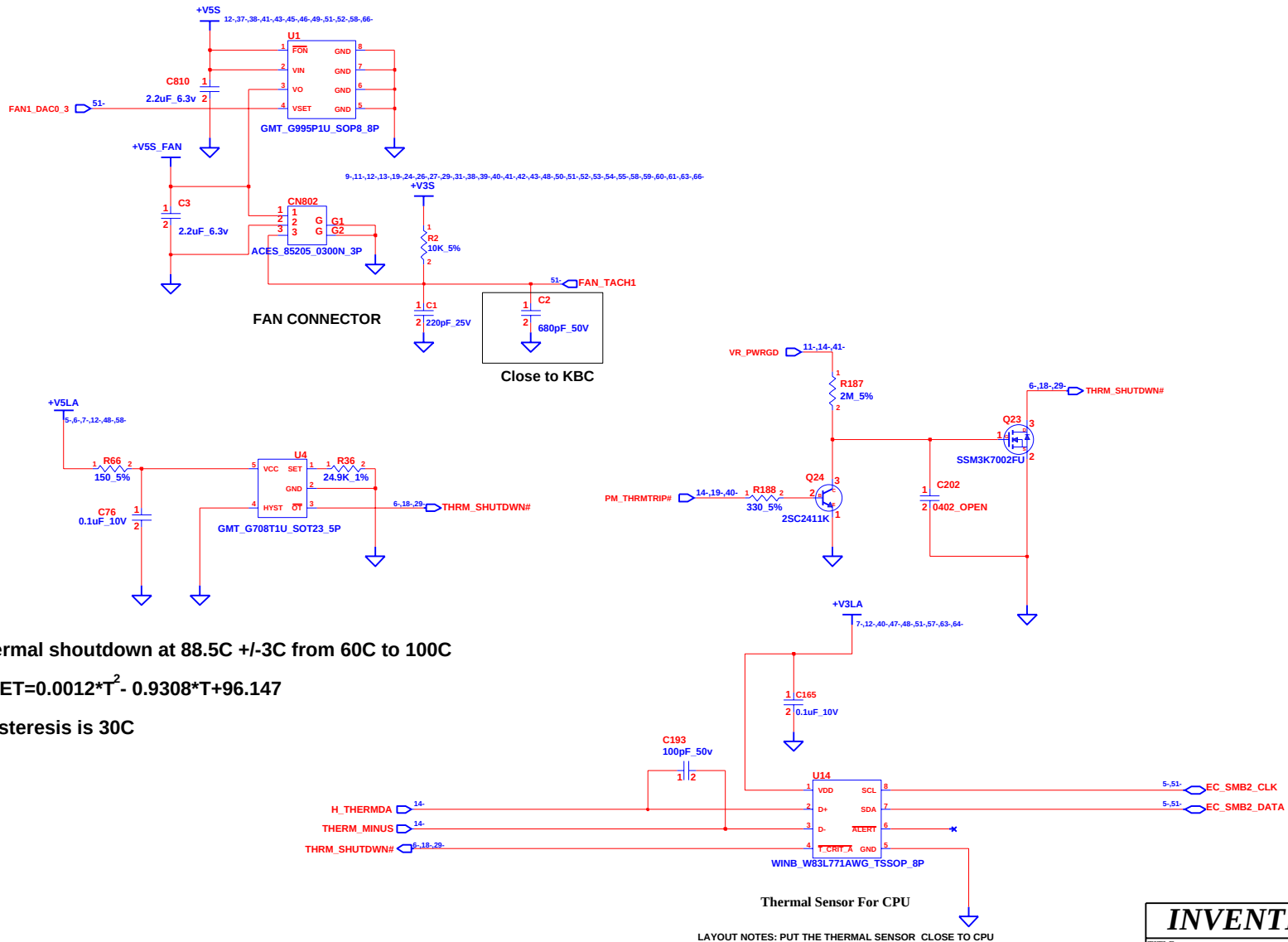
AF25

VSS162

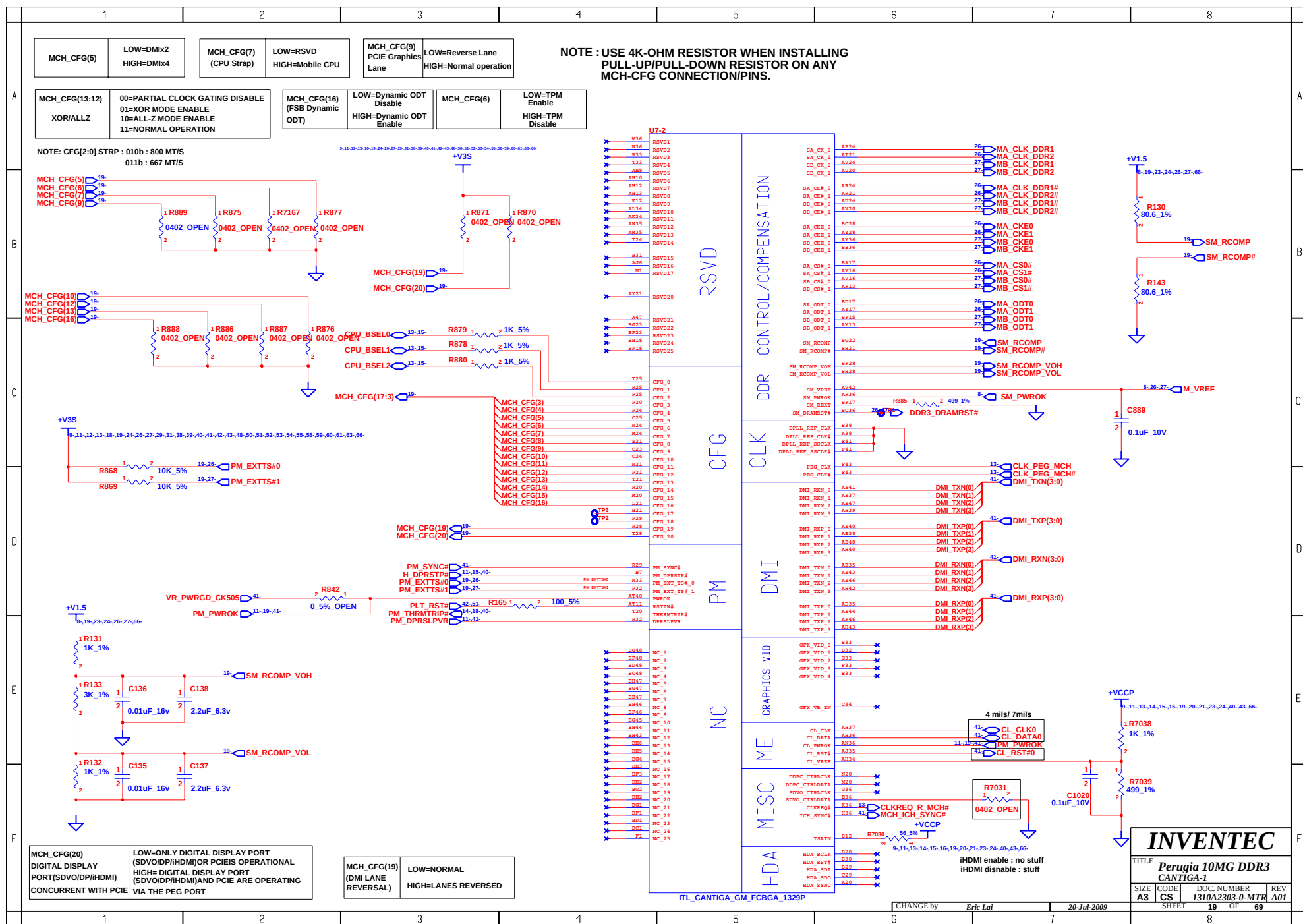
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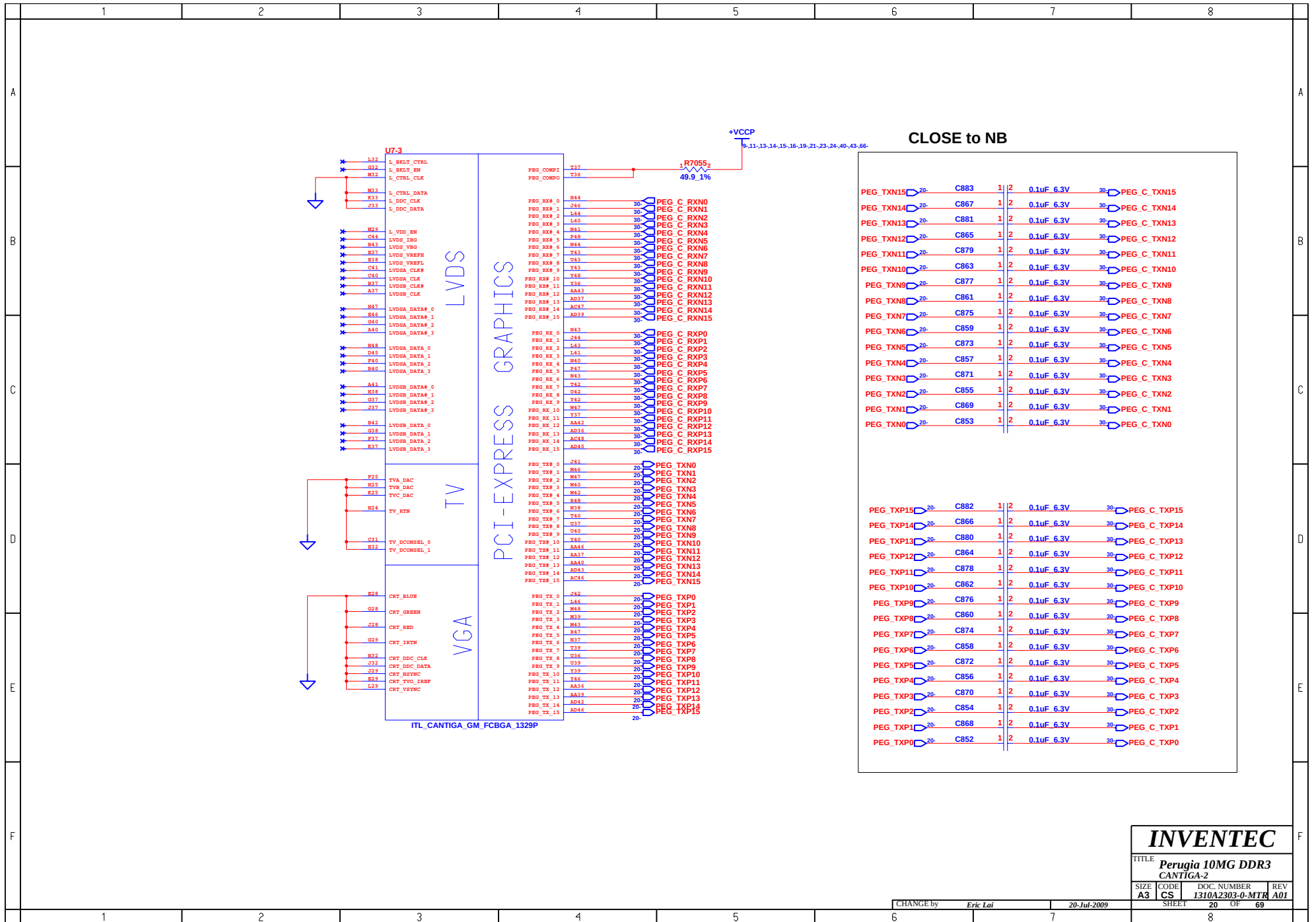
VSS163

FOX_PZ4782K_274M_41_478P



INVENTEC			
TITLE			
Perugia 10MG DDR3			
THERMAL&FAN CONTROLLER			
SIZE	CODE	DOC. NUMBER	REV
A3	CS	I310AZ303-0-MTR	A01
CHANGE by		20-Jul-2009	
Eric Lai			
SHEET		18	OF 69





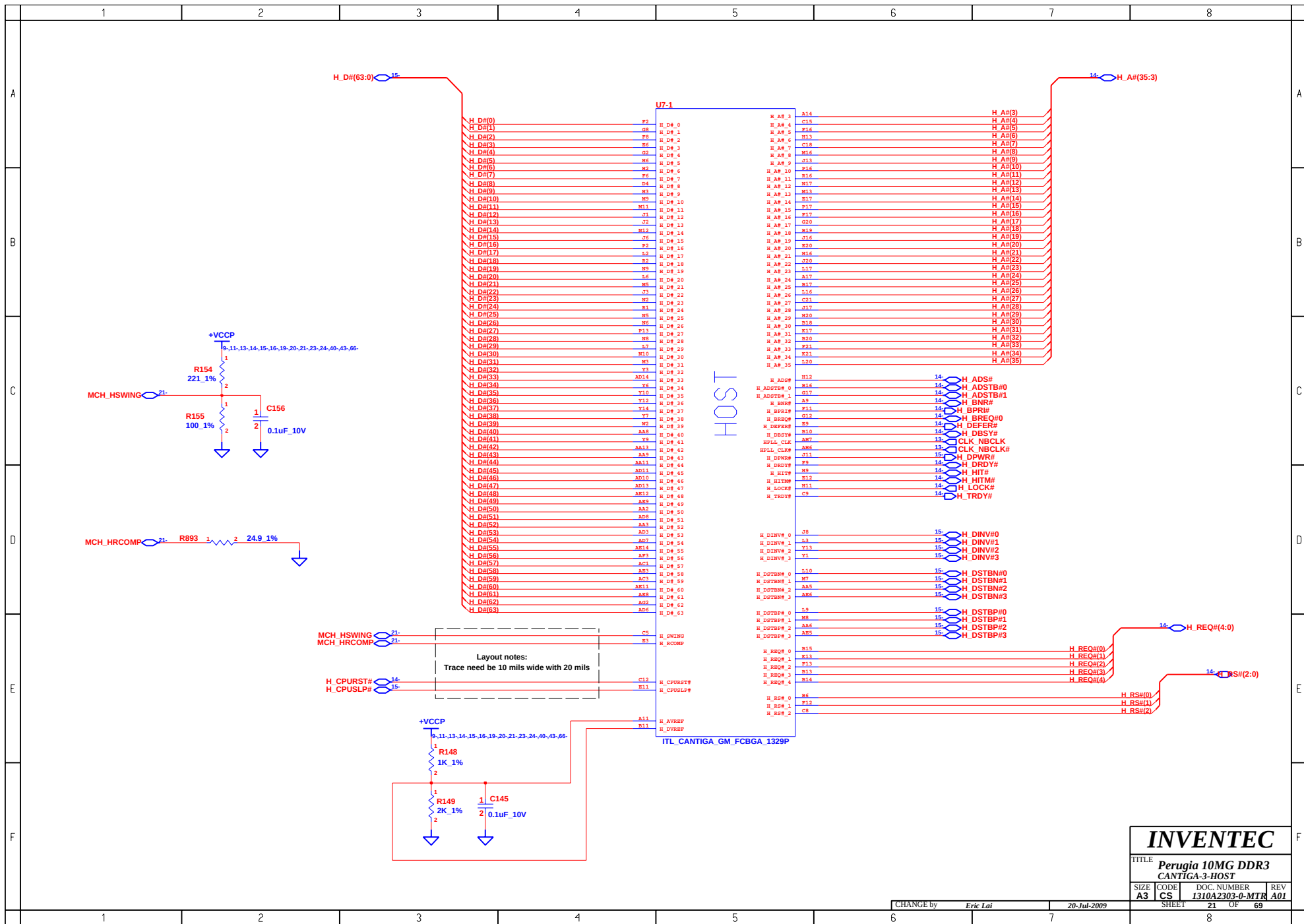
INVENTEC

TITLE
Perugia 10MG DDR3
CANTIGA-2

SIZE	CODE	DOC. NUMBER	REV
A3	CS	1310A2303-0-MTR	A01

CHANGE by Eric Lai 20-Jul-2009

SHEET 20 OF 69



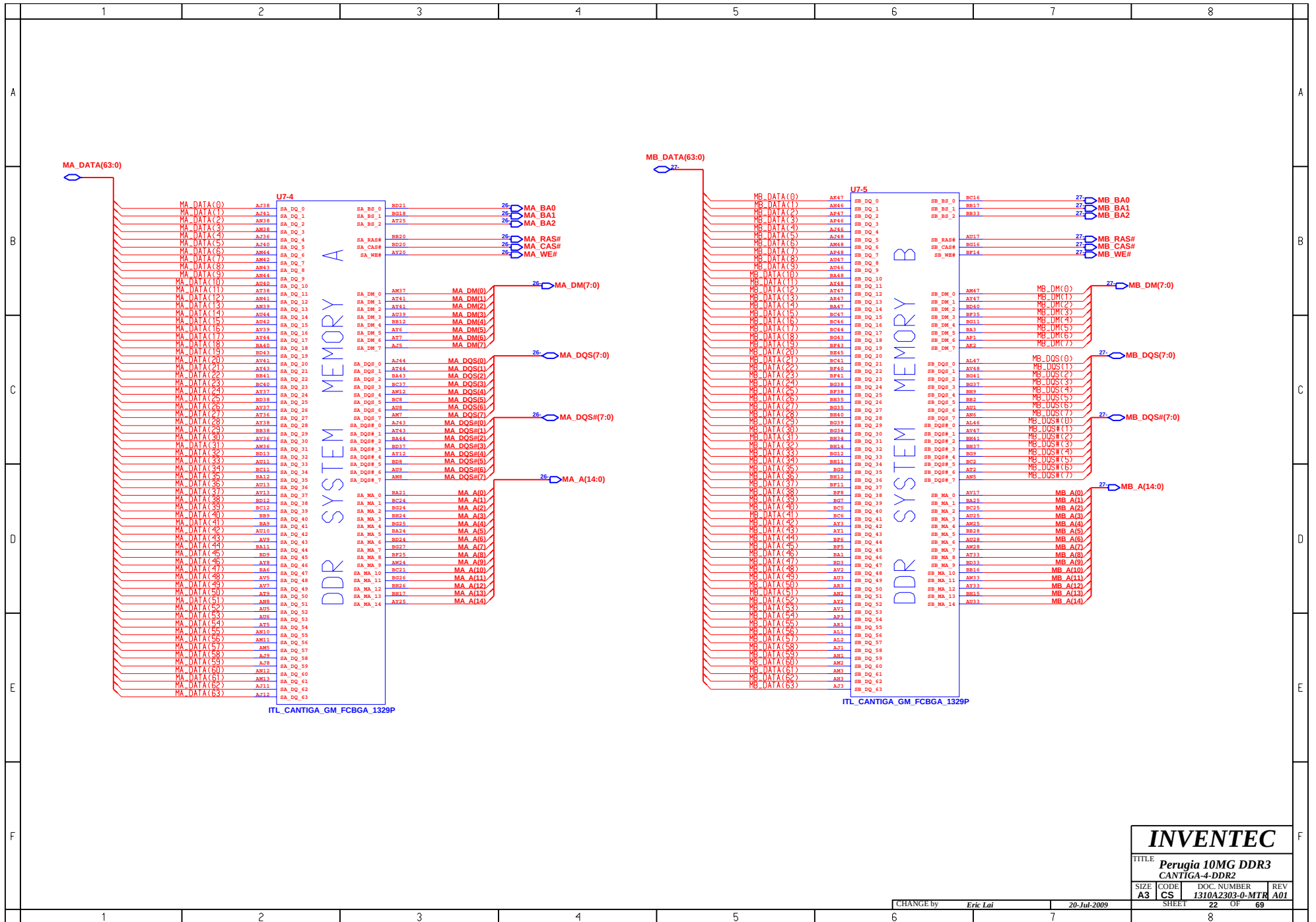
INVENTEC

TITLE
**Perugia 10MG DDR3
CANTIGA-3-HOST**

SIZE	CODE	DOC. NUMBER	REV
A3	CS	1310AZ303-0-MTR	A01

CHANGE by Eric Lai 20-Jul-2009

SHEET 21 OF 69



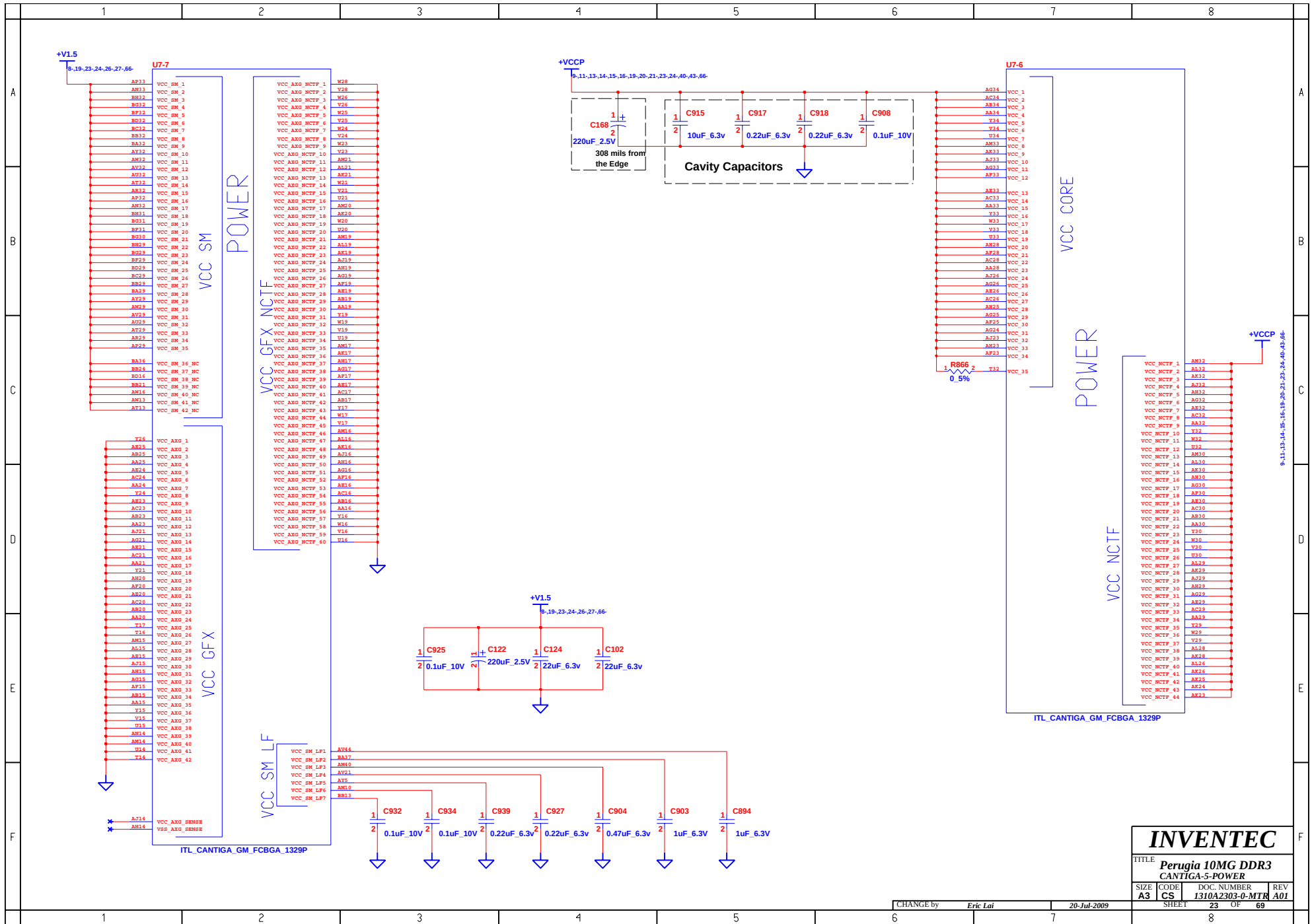
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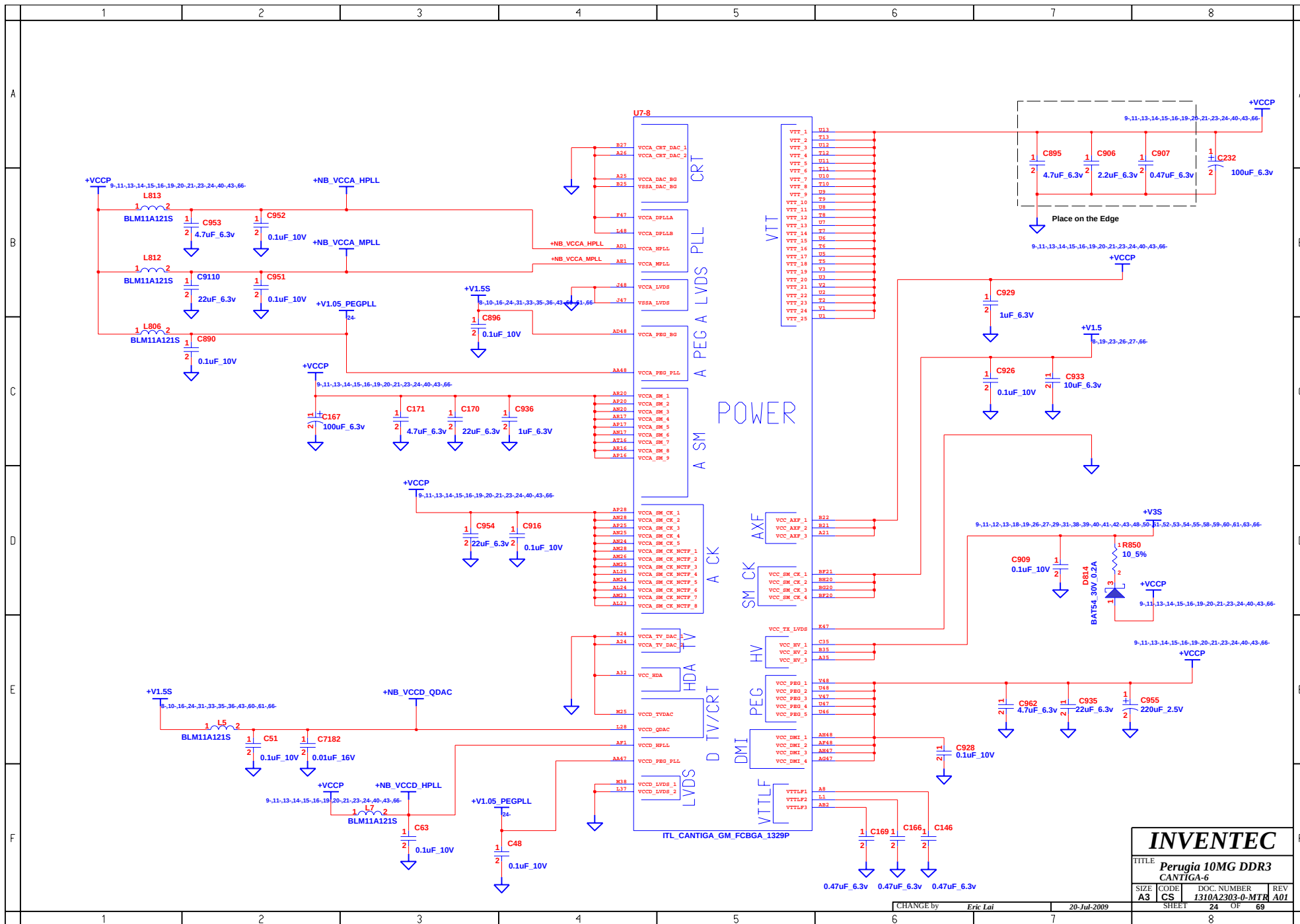
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**Perugia 10MG DDR3
CANTIGA-4-DDR2**

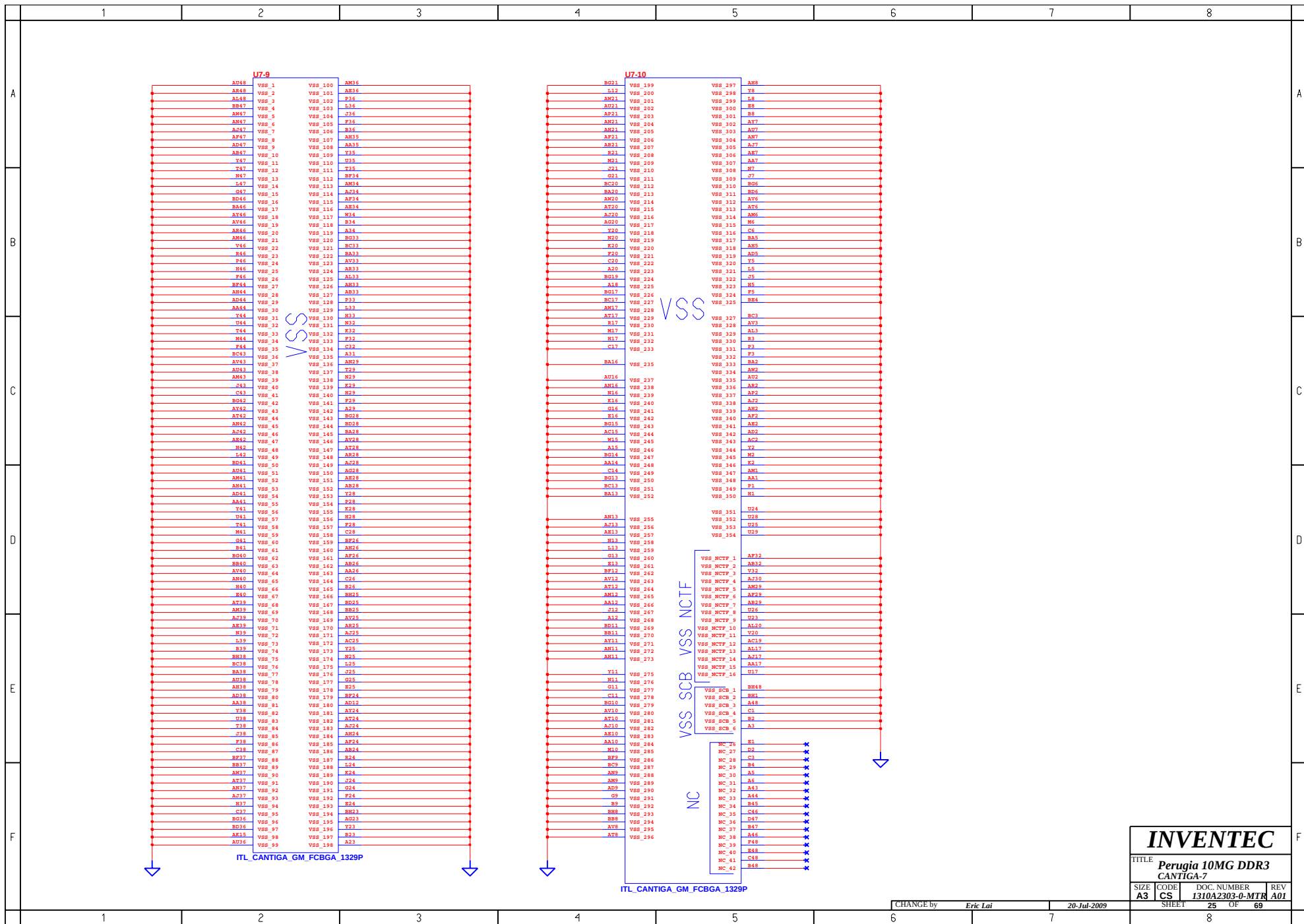
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A3	CS	1310AZ303-0-MTR	A01

CHANGE by **Eric Lai** 20-Jul-2009

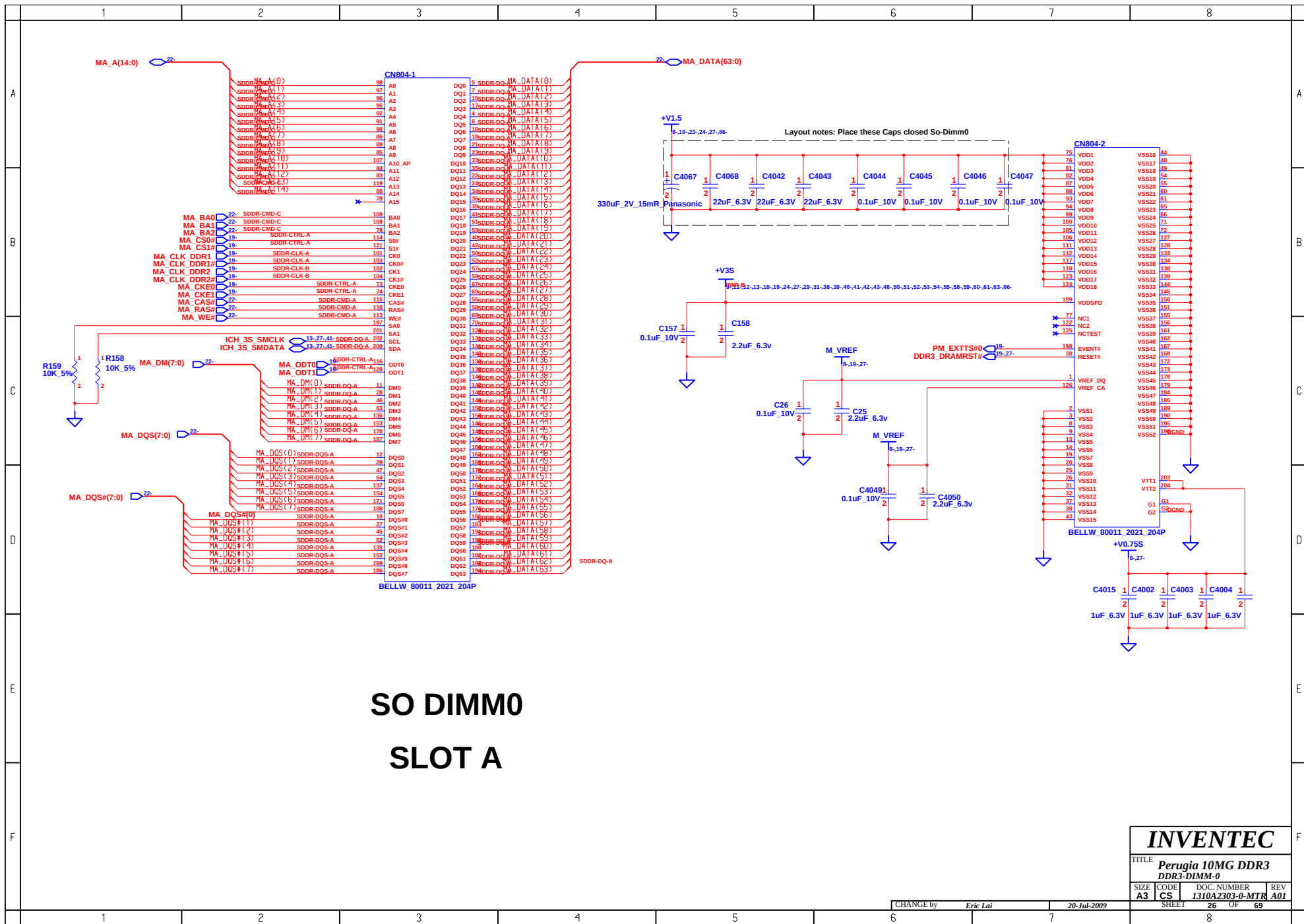
SHEET 22 OF 69

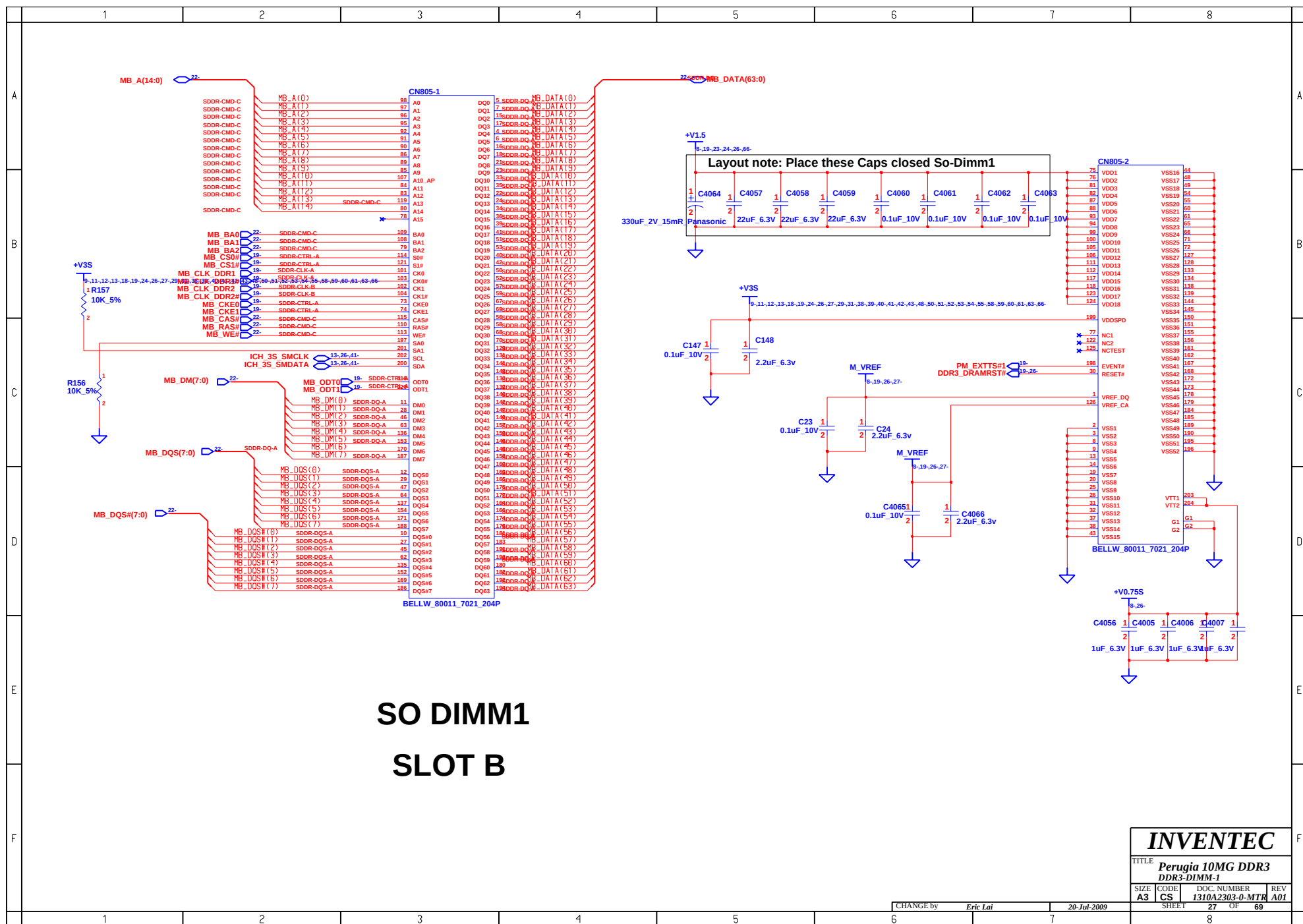






INVENTEC			
TITLE Perugia 10MG DDR3 CANTIGA-7			
SIZE A3	CODE CS	DOC. NUMBER I310AZ303-0-MTR	REV A01
SHEET		25	OF 69





	1	2	3	4	5	6	7	8	
A									A
B									B
C									C
D									D
E									E
F									F
						CHANGE by <i>Eric Lal</i>		27-Sep-2003	
1	2	3	4	5	6	7	8		

INVENTEC

TITLE **Perugia 10MG DDR3**
DDR2-DAMPING

SIZE	CODE	DOC. NUMBER	REV
A3	CS	1310AZ303-0-MTR	A01

SHEET 28 OF 69

M92 XT P/N = 6019B0591602

MEM_ID0	MEM_ID1	MEM_ID2	MEM_ID3	MEM_ID4
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0

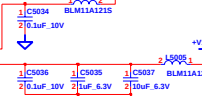
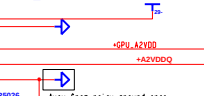
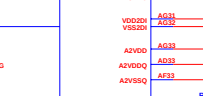
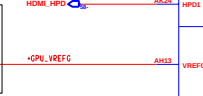
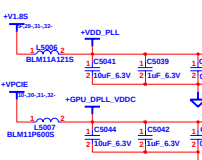
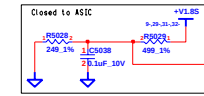
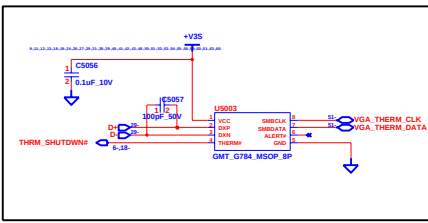
MEM_ID0	MEM_ID1	MEM_ID2	MEM_ID3	MEM_ID4
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0

GPI0_00	GPI0_01	GPI0_02	Memory aperture size
0	0	0	256M
0	0	0	512M
0	0	0	1024M

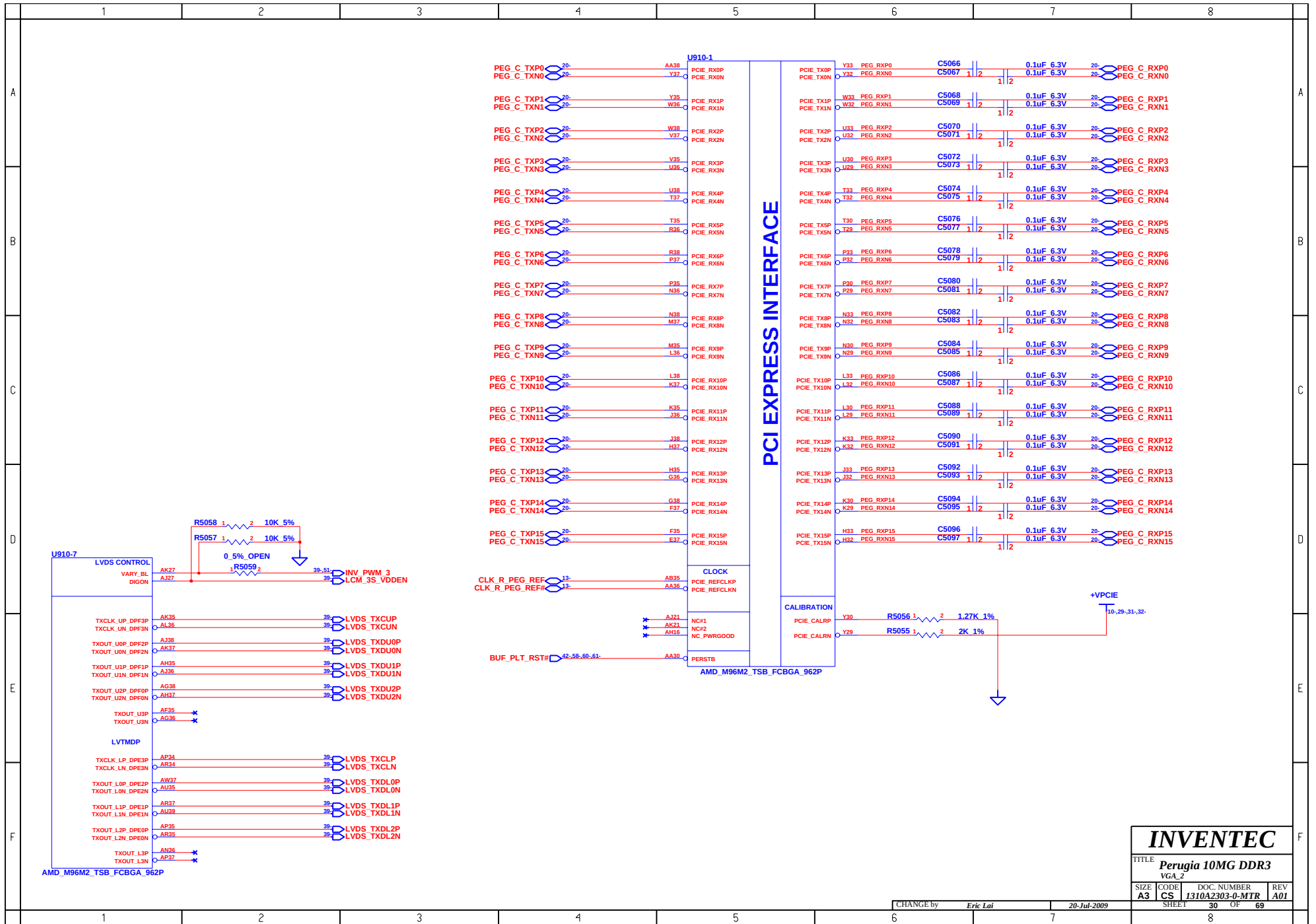
MEM_ID0	VRAM Vendor & Size
00h	Hynix 6600M (256M)
01h	SanDisk 6600M (256M)
02h	SanDisk 6600M (512M)
03h	SanDisk 6600M (1024M)
04h	SanDisk 6600M (2048M)
05h	SanDisk 6600M (4096M)
06h	SanDisk 6600M (8192M)
07h	SanDisk 6600M (16384M)

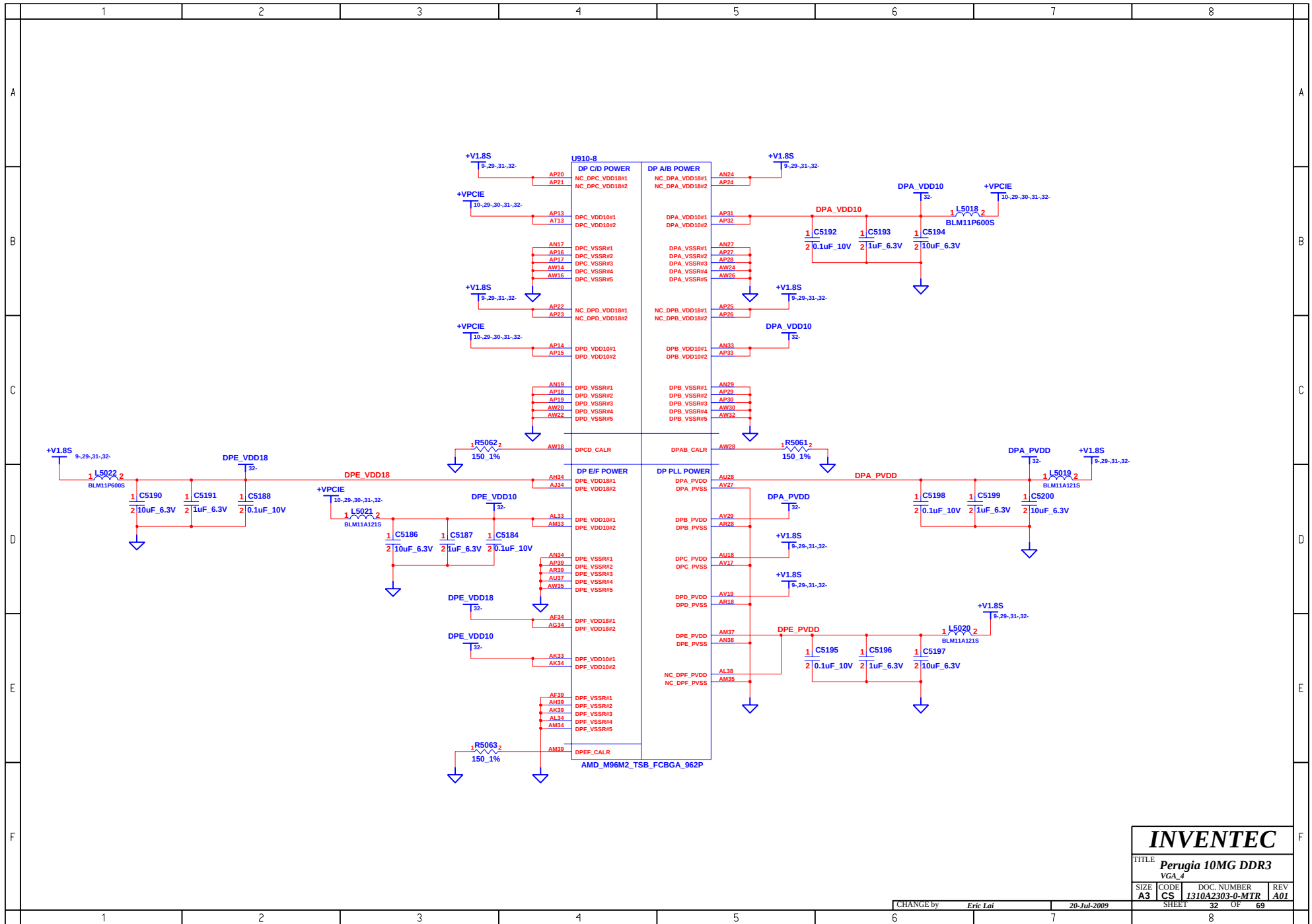
Place close to ASIC

CRT_D_VSYNC - R5051 1 2 10K 5%
 CRT_S - R5043 1 2 10K 1%
 CRT_R - R5044 1 2 10K 1%



Engineer	Eric Lai
Drawn by	Eric Lai
R&D CHK	
DOC CTRL CHK	
MFG CTRL CHK	
QA CHK	
REV	A01
Drawing Number	Sheet 29 of 69



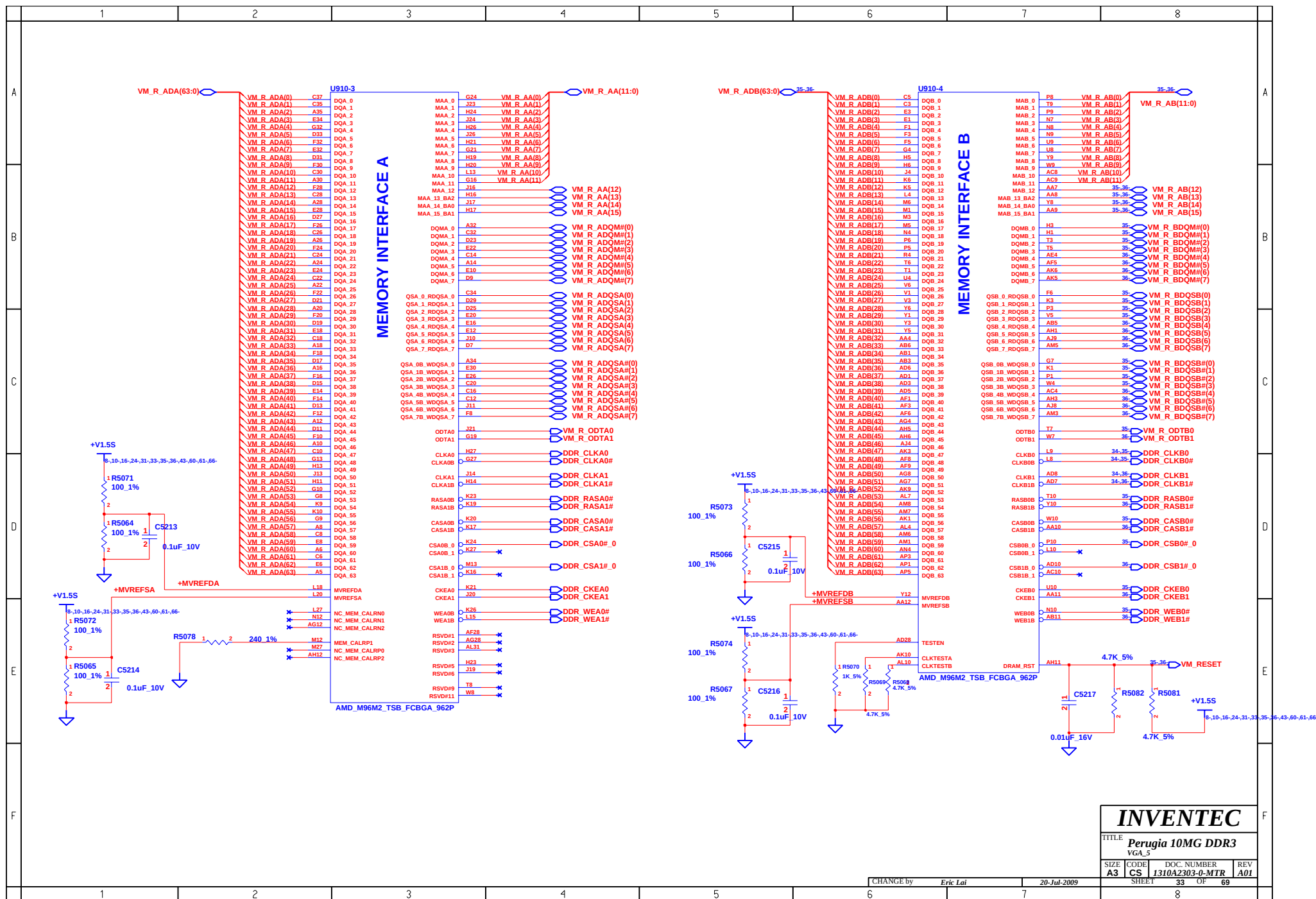


INVENTEC

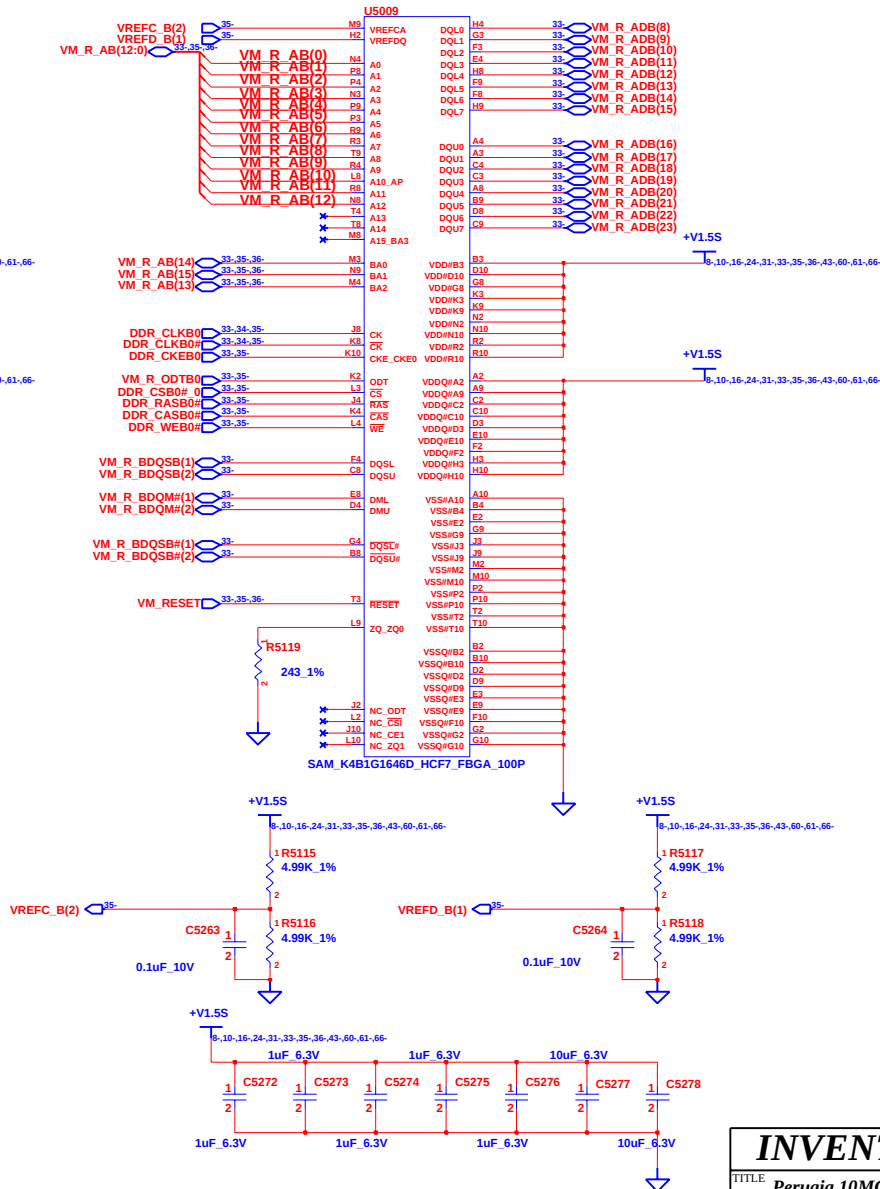
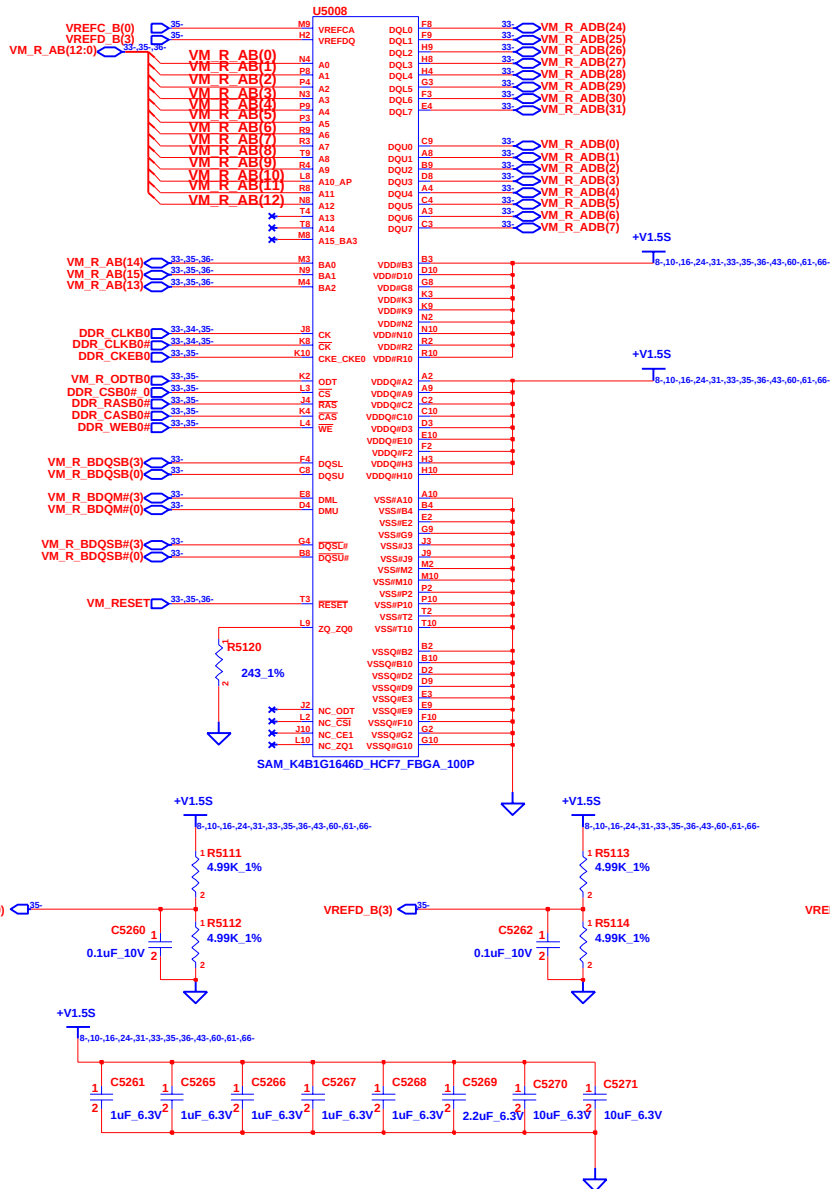
TITLE			
VGA_4			
Perugia 10MG DDR3			
SIZE	CODE	DOC. NUMBER	REV
A3	CS	1310A2303-0-MTR	A01

CHANGE by Eric Lai 20-Jul-2009

SHEET 32 OF 69



M92 Use channel B



INVENTEC

TITLE
Perugia 10MG DDR3

SIZE CODE DOC. NUMBER REV
A3 CS 1310A2303-0-MTR A01

SHEET 35 OF 69

CHANGE by Eric Lai 20-Jul-2009

M92 Use channel B

U5010

VREFC B(4) 36-
VREFD B(7) 36-
VM_R_AB(12:0) 33-35-36-

VM R AB(0) 33-
VM R AB(1) 35-
VM R AB(2) 36-
VM R AB(3) 33-
VM R AB(4) 35-
VM R AB(5) 36-
VM R AB(6) 33-
VM R AB(7) 35-
VM R AB(8) 36-
VM R AB(9) 33-
VM R AB(10) 35-
VM R AB(11) 36-
VM R AB(12) 33-35-36-

VM R AB(14) 33-35-36-
VM R AB(15) 33-35-36-
VM_R_AB(13) 33-35-36-

DDR_CLKB1 33-34-35-
DDR_CLKB1# 33-34-35-
DDR_CKEB1 33-35-36-

VM R ODTB1 33-35-36-
DDR_CSB1# 0 33-35-36-
DDR_RASB1# 33-35-36-
DDR_CASB1# 33-35-36-
DDR_WEB1# 33-35-36-

VM R BDQSB(6) 33-
VM_R_BDQSB(7) 33-
VM R BQDM(6) 33-
VM_R_BQDM(7) 33-
VM R BDQSB(6) 33-
VM_R_BDQSB(7) 33-
VM_RESET 33-35-36-

RS130 243 1%
NC_ODT 32
NC_CSI 32
NC_CEI 32
NC_ZQ1 32

SAM_K4B1G1646D_HCF7_FBGA_100P

U5011

VREFC B(6) 36-
VREFD B(5) 36-
VM_R_AB(12:0) 33-35-36-

VM R AB(0) 33-
VM R AB(1) 35-
VM R AB(2) 36-
VM R AB(3) 33-
VM R AB(4) 35-
VM R AB(5) 36-
VM R AB(6) 33-
VM R AB(7) 35-
VM R AB(8) 36-
VM R AB(9) 33-
VM R AB(10) 35-
VM R AB(11) 36-
VM_R_AB(12) 33-35-36-

VM R AB(14) 33-35-36-
VM R AB(15) 33-35-36-
VM_R_AB(13) 33-35-36-

DDR_CLKB1 33-34-35-
DDR_CLKB1# 33-34-35-
DDR_CKEB1 33-35-36-

VM R ODTB1 33-35-36-
DDR_CSB1# 0 33-35-36-
DDR_RASB1# 33-35-36-
DDR_CASB1# 33-35-36-
DDR_WEB1# 33-35-36-

VM R BDQSB(4) 33-
VM_R_BDQSB(5) 33-
VM R BQDM(4) 33-
VM_R_BQDM(5) 33-
VM R BDQSB(4) 33-
VM_R_BDQSB(5) 33-
VM_RESET 33-35-36-

RS129 243 1%
NC_ODT 32
NC_CSI 32
NC_CEI 32
NC_ZQ1 32

SAM_K4B1G1646D_HCF7_FBGA_100P

Power and Ground Planes

+V1.5S 8-10-16-24-31-33-35-36-43-60-61-66-

0.1uF_10V

1uF_6.3v

2.2uF_6.3v

10uF_6.3v

Components

RS120 4.99K 1%
RS121 4.99K 1%
RS122 4.99K 1%
RS123 4.99K 1%
RS124 4.99K 1%
RS125 4.99K 1%
RS126 4.99K 1%
RS127 4.99K 1%
RS128 4.99K 1%

C5279 0.1uF_10V
C5280 1uF_6.3v
C5281 1uF_6.3v
C5282 1uF_6.3v
C5283 1uF_6.3v
C5284 1uF_6.3v
C5285 1uF_6.3v
C5286 1uF_6.3v
C5287 1uF_6.3v
C5288 1uF_6.3v
C5289 1uF_6.3v
C5290 1uF_6.3v
C5291 1uF_6.3v
C5292 1uF_6.3v
C5293 1uF_6.3v
C5294 1uF_6.3v
C5295 1uF_6.3v
C5296 1uF_6.3v
C5297 1uF_6.3v

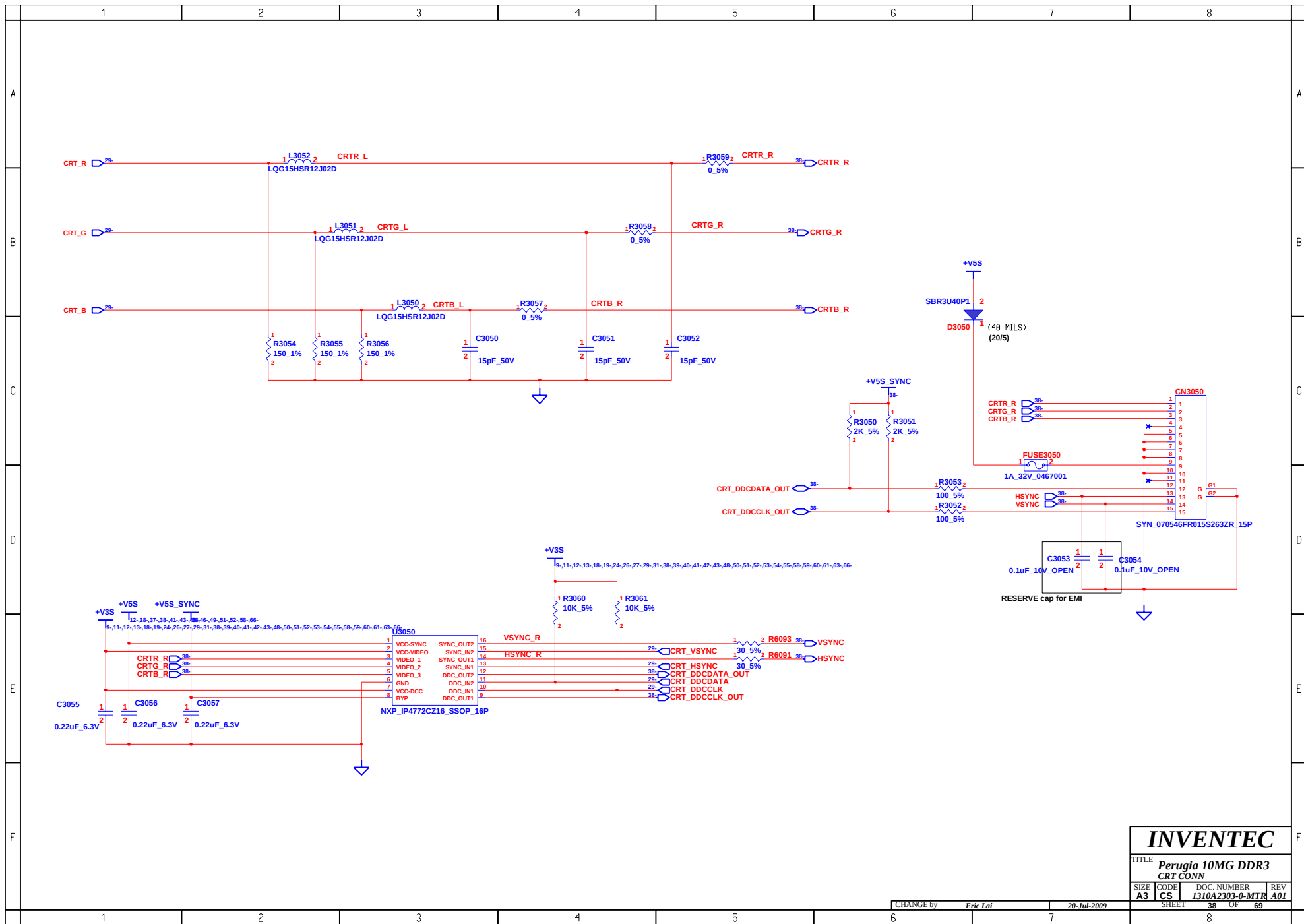
INVENTEC

TITLE Perugia 10MG DDR3
VRAM_1

SIZE A3 CODE CS DOC. NUMBER 1310A2303-0-MTR REV A01

SHEET 36 OF 69

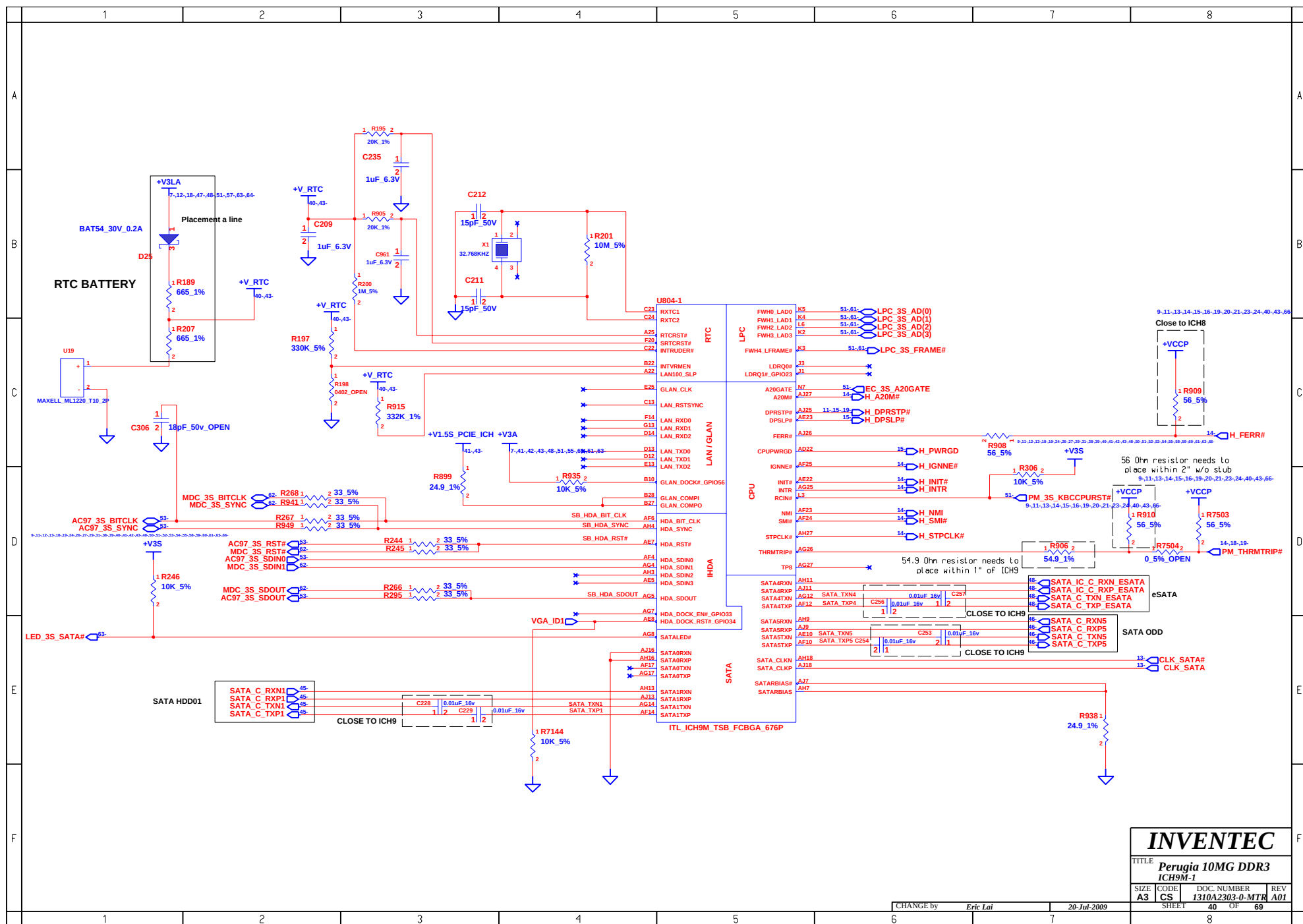
CHANGE by Eric Lai 20-Jul-2009

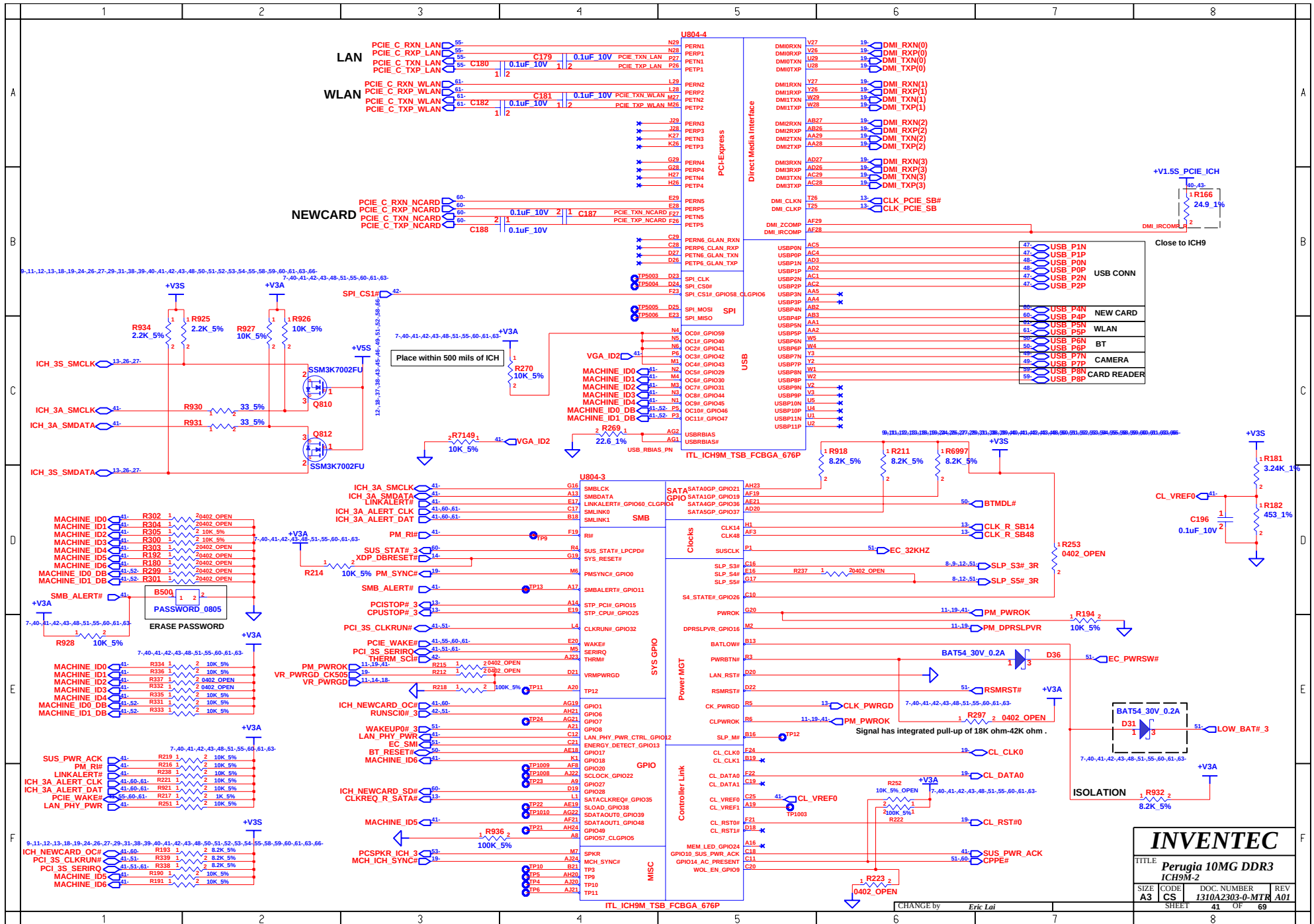


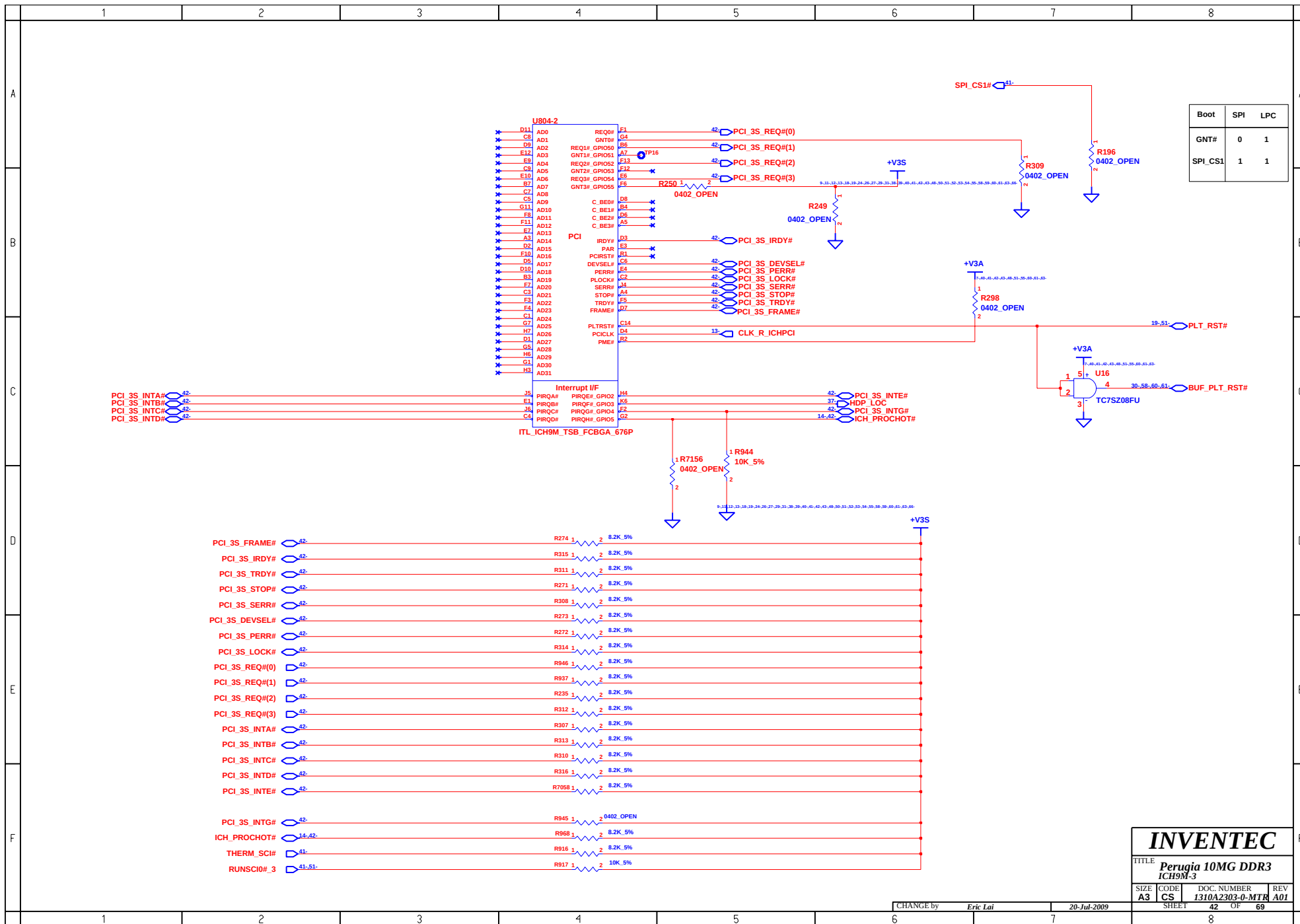
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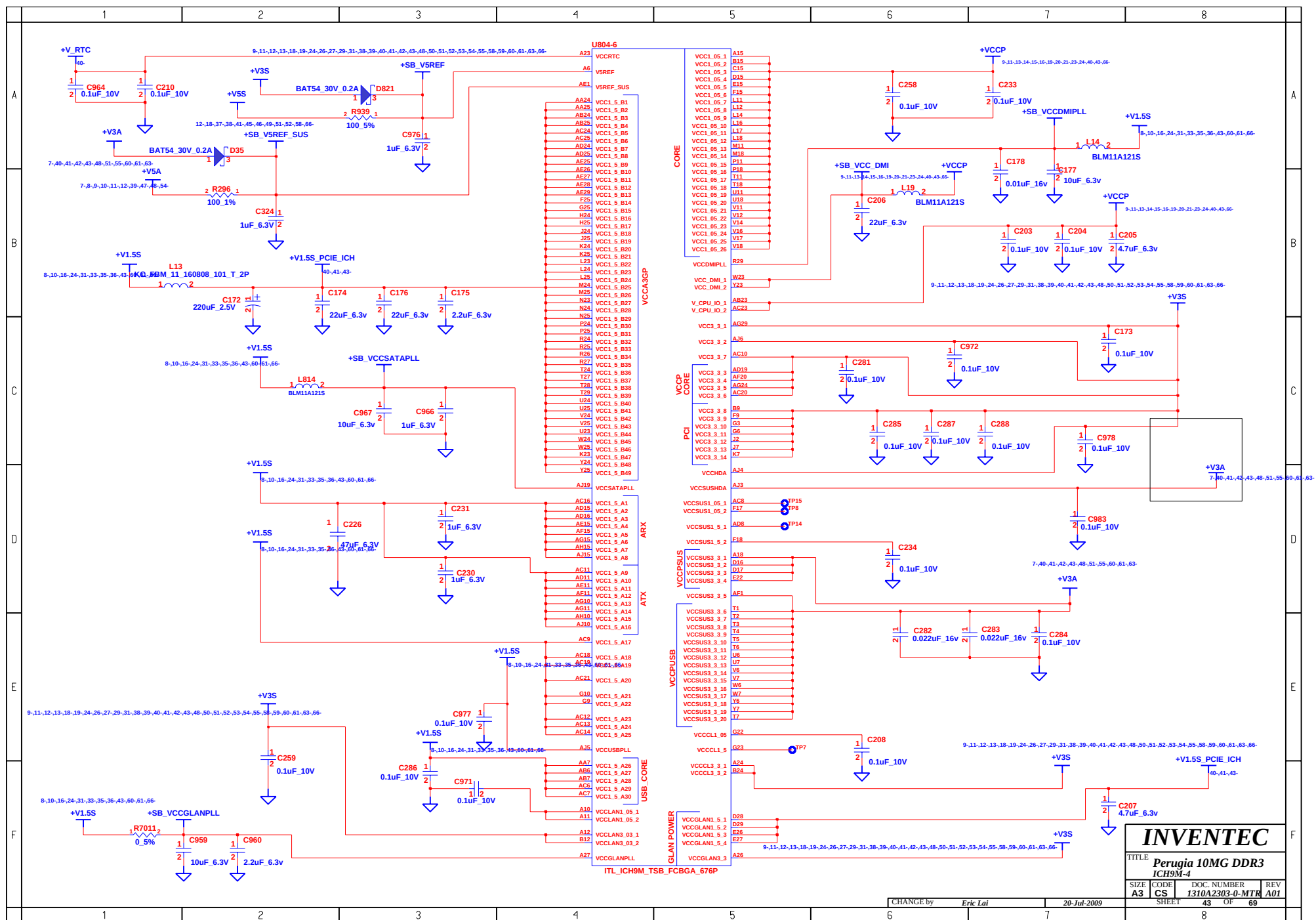
TITLE			
Perugia 10MG DDR3			
CRT CONN			
SIZE	CODE	DOC. NUMBER	REV
A3	CS	I310AZ303-0-MTR	A01
SHEET		38	OF 69

CHANGE by Eric Lai 20-Jul-2009



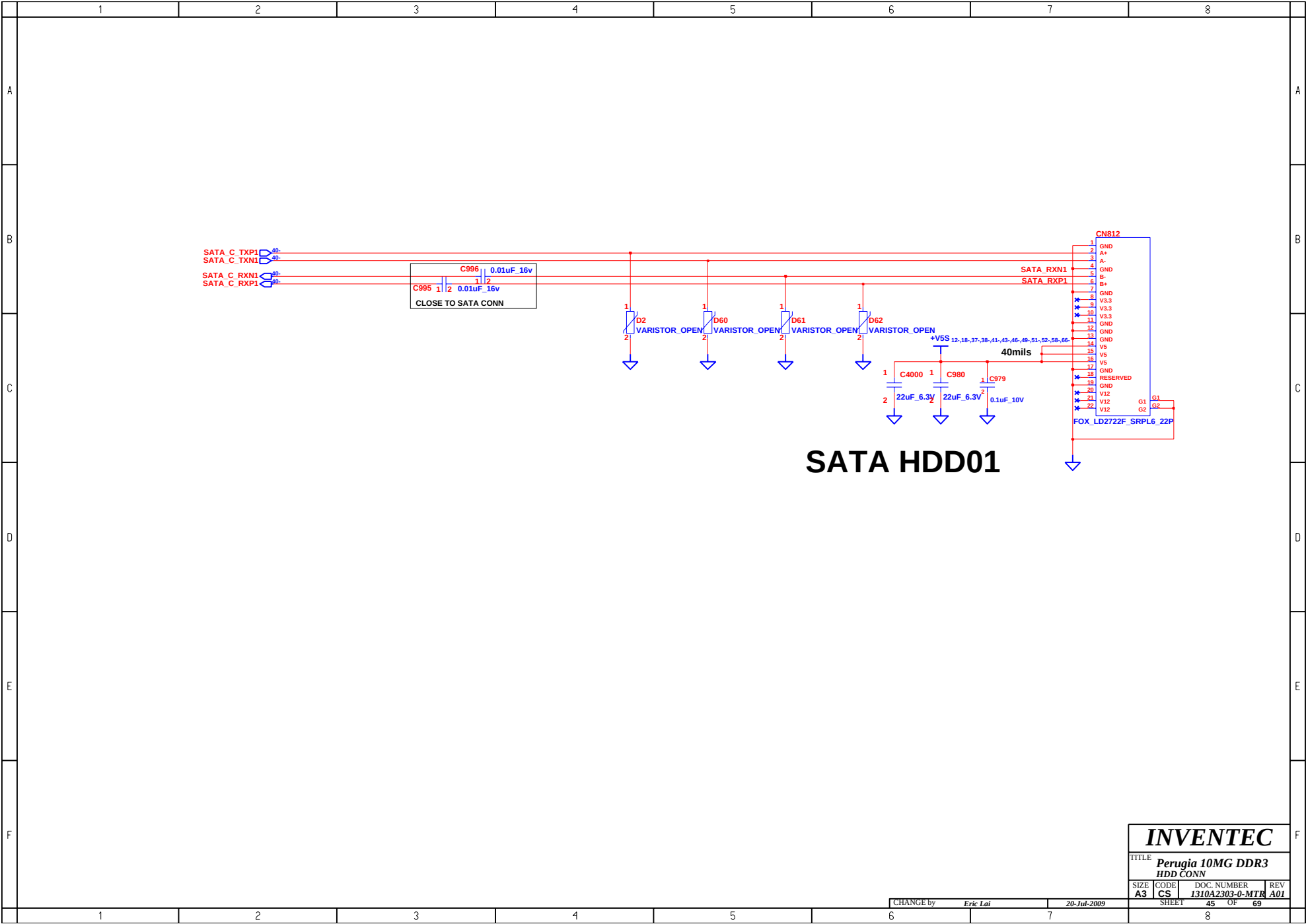






	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

INVENTEC			
TITLE Perugia 10MG DDR3 ICH9M-5			
SIZE A3	CODE CS	DOC. NUMBER 1310A2303-0-MTR	REV A01
CHANGE by Eric Lai		20-Jul-2009	
SHEET 44		OF 69	



SATA HDD01

INVENTEC

TITLE
Perugia 10MG DDR3
HDD CONN

SIZE A3	CODE CS	DOC. NUMBER I310AZ303-0-MTR	REV A01
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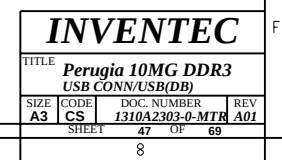
CHANGE by Eric Lai 20-Jul-2009

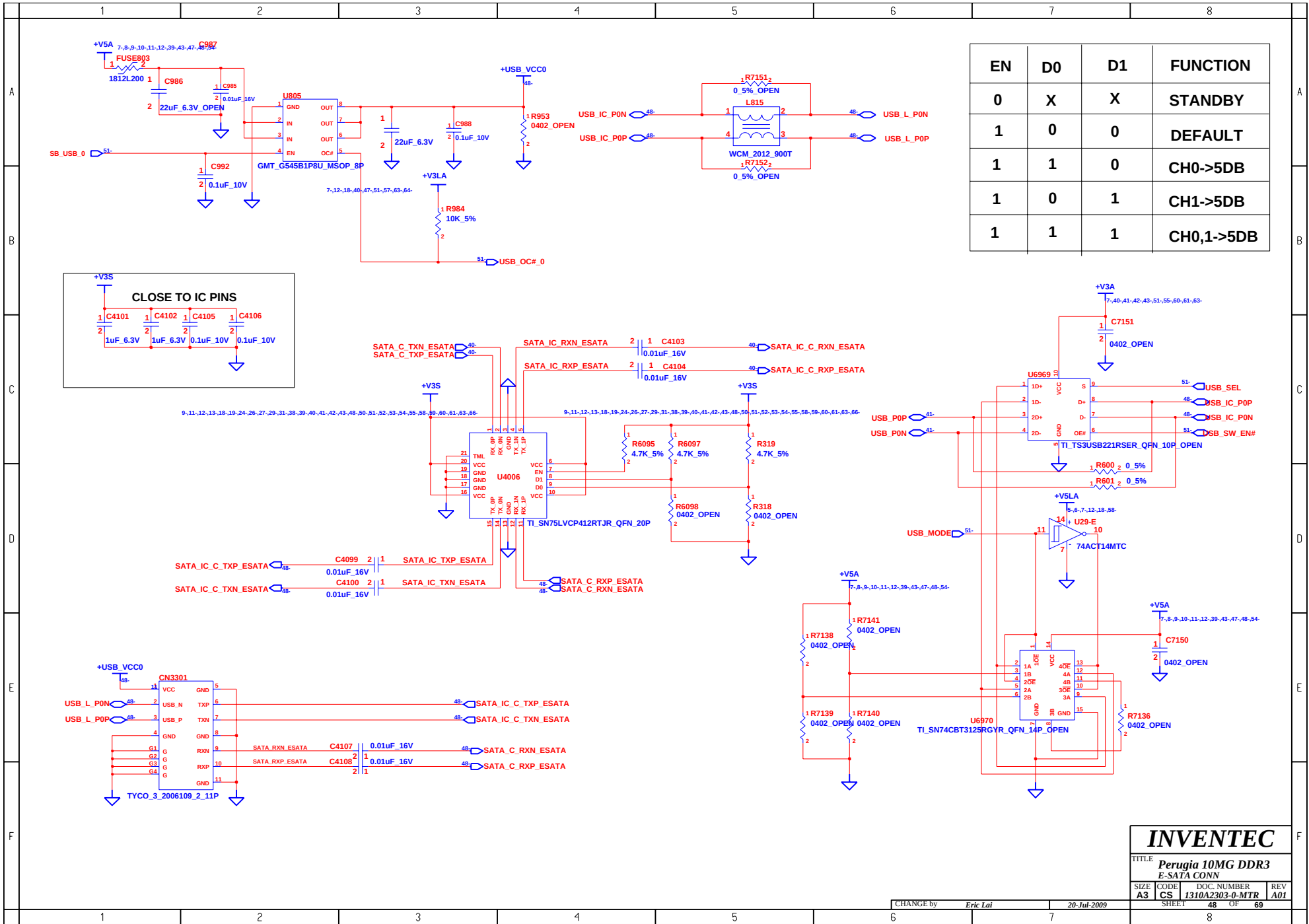
SHEET 45 OF 69



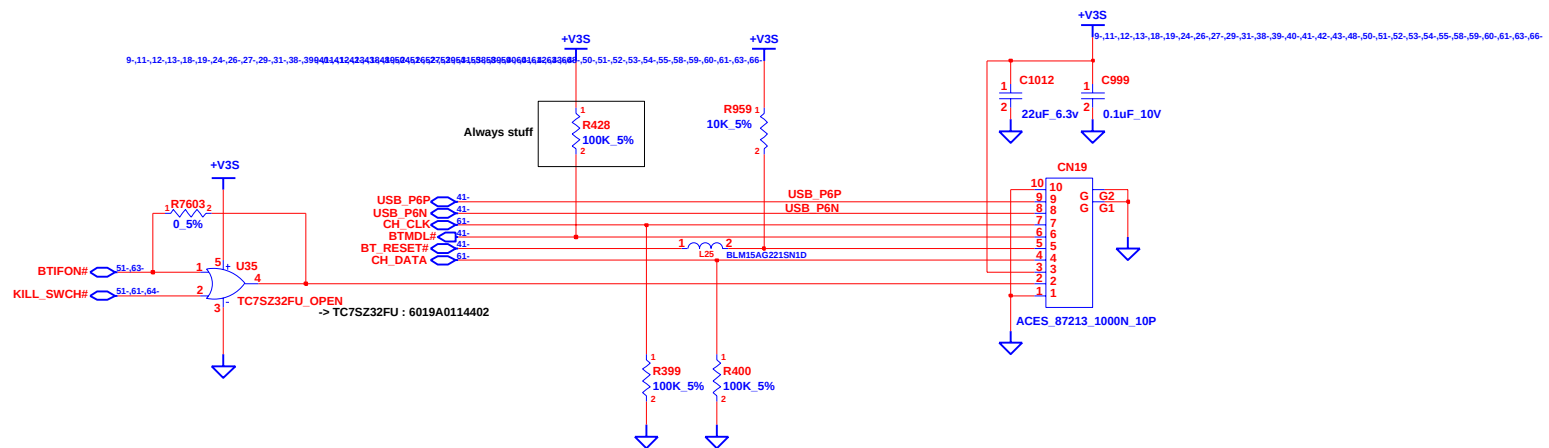
TITLE	Perugia 10MG DDR3 SATA ODD
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CHANGE by	<i>Eric Lai</i>	20-Jul-2009
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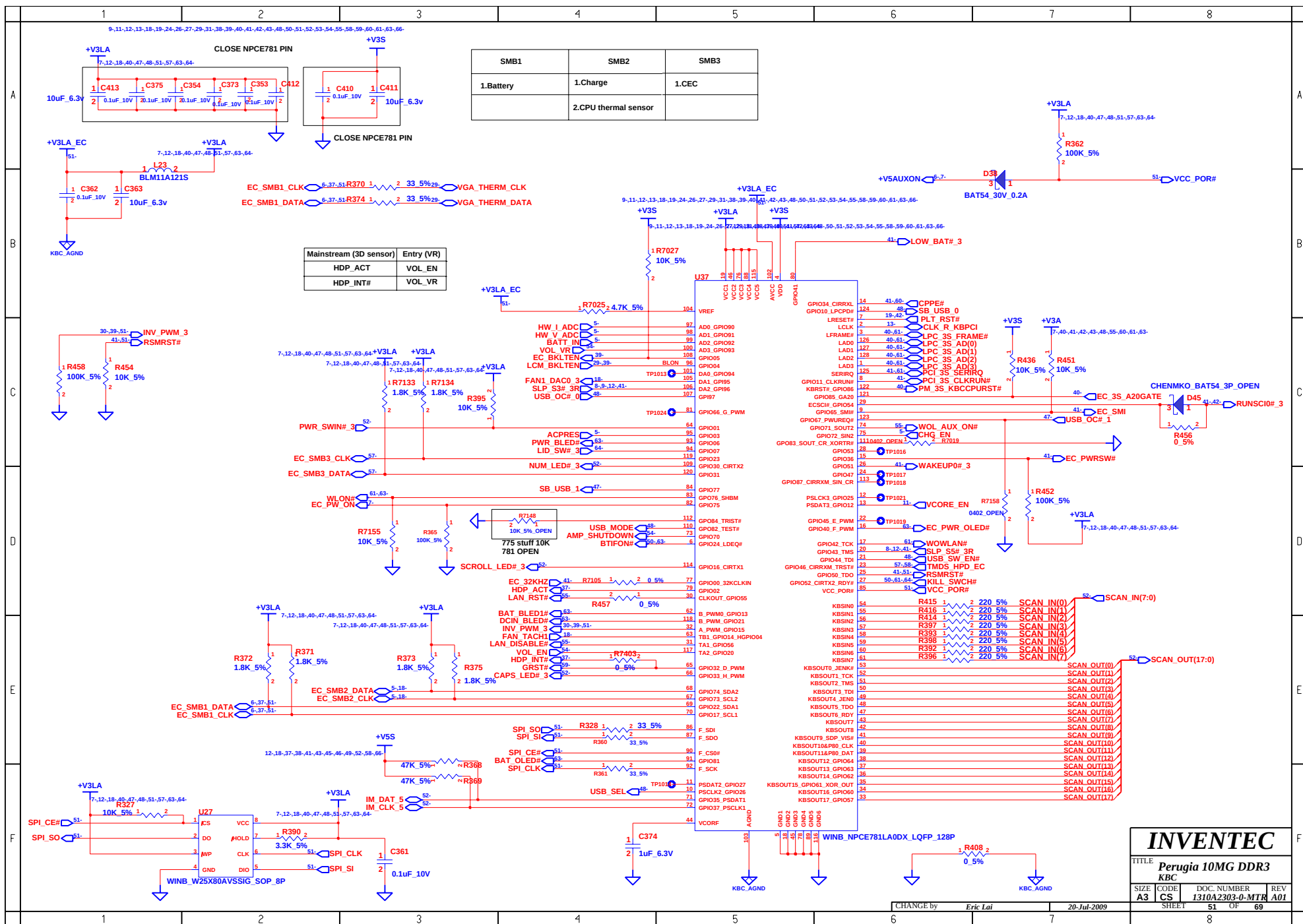
INVENTEC

TITLE
**Perugia 10MG DDR3
Bluetooth**

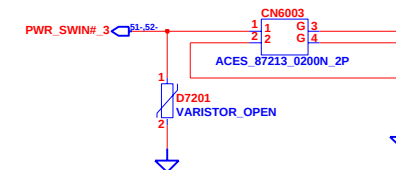
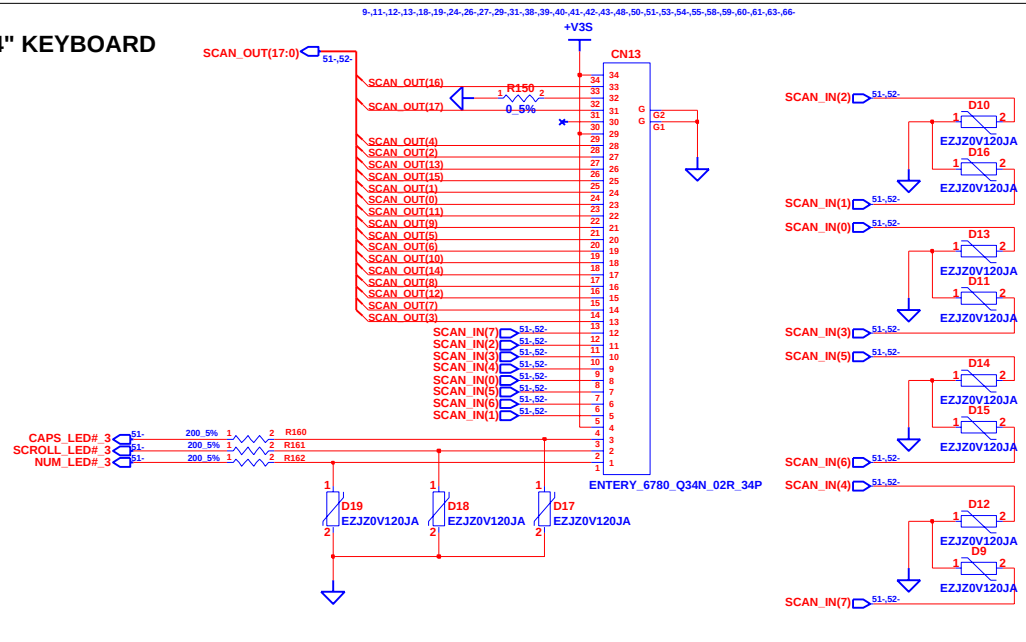
SIZE A3	CODE CS	DOC. NUMBER 1310A2303-0-MTR	REV A01
SHEET 50		OF 69	

CHANGE by *Eric Lai* 20-Jul-2009

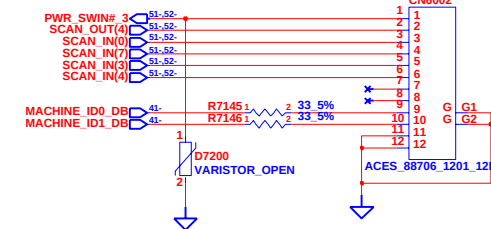
8



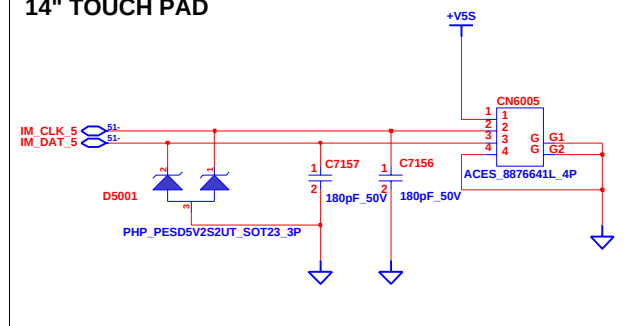
14" KEYBOARD



Hotkey



14" TOUCH PAD



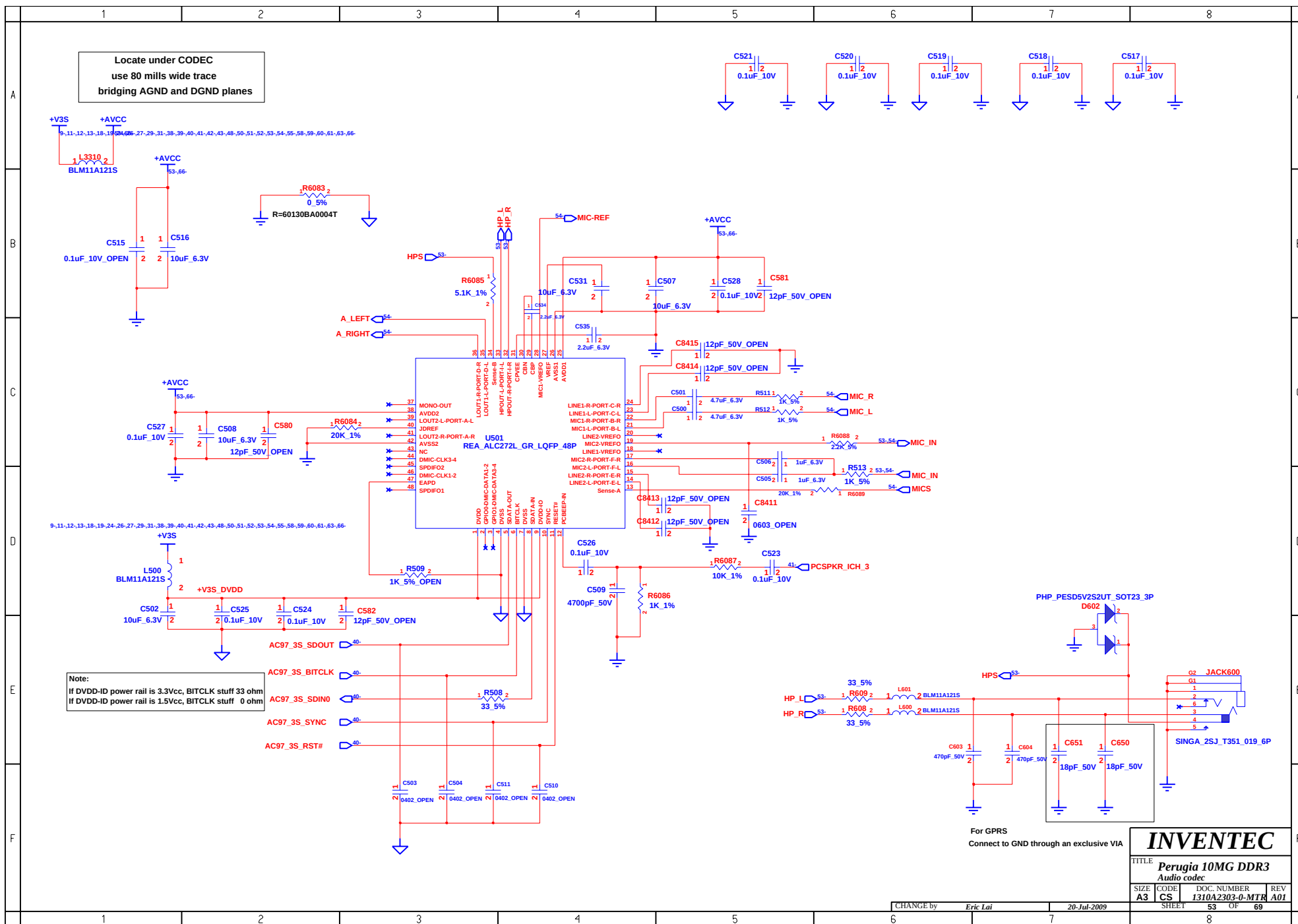
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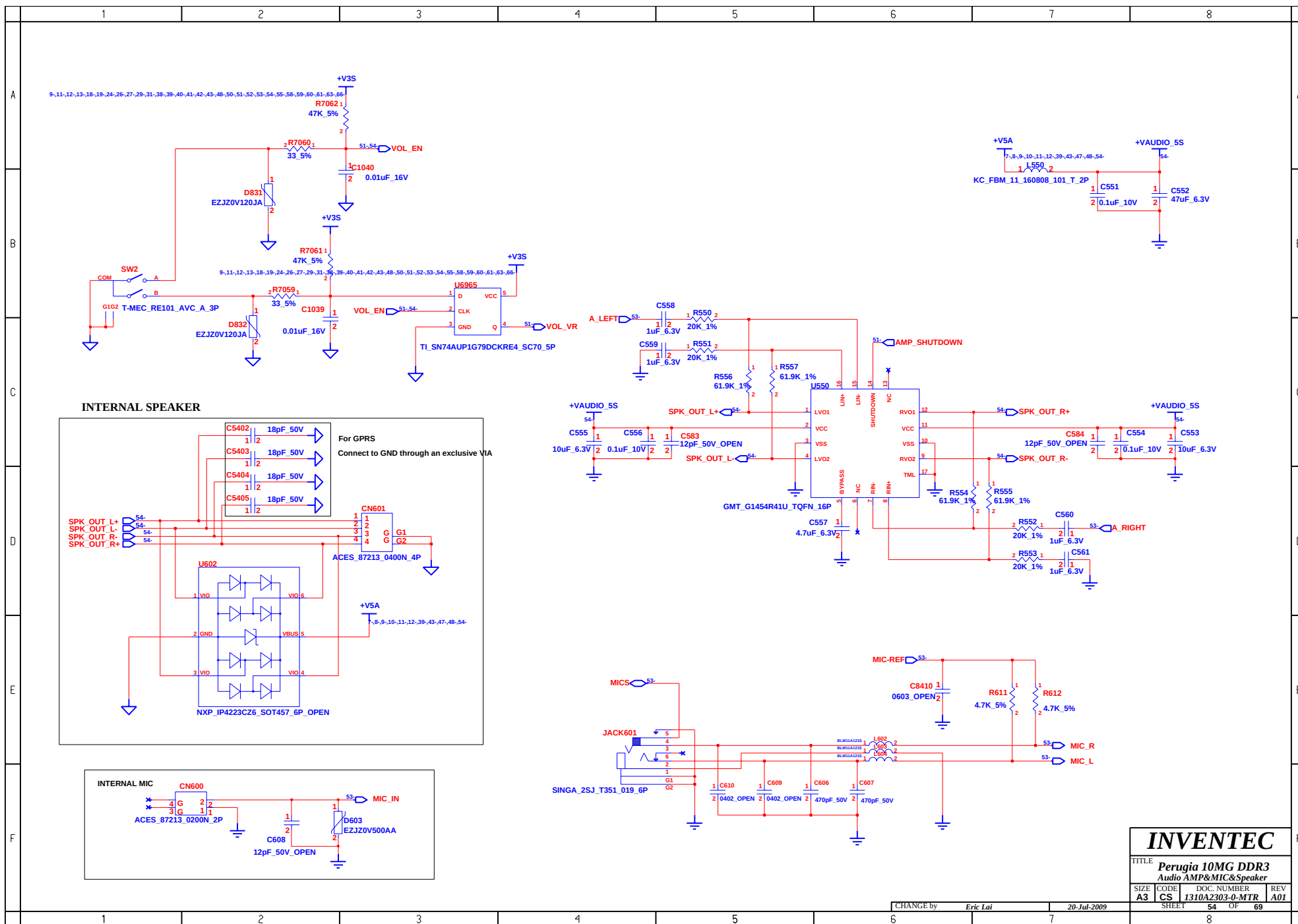
TITLE
**Perugia 10MG DDR3
K/B & TP/B CONN**

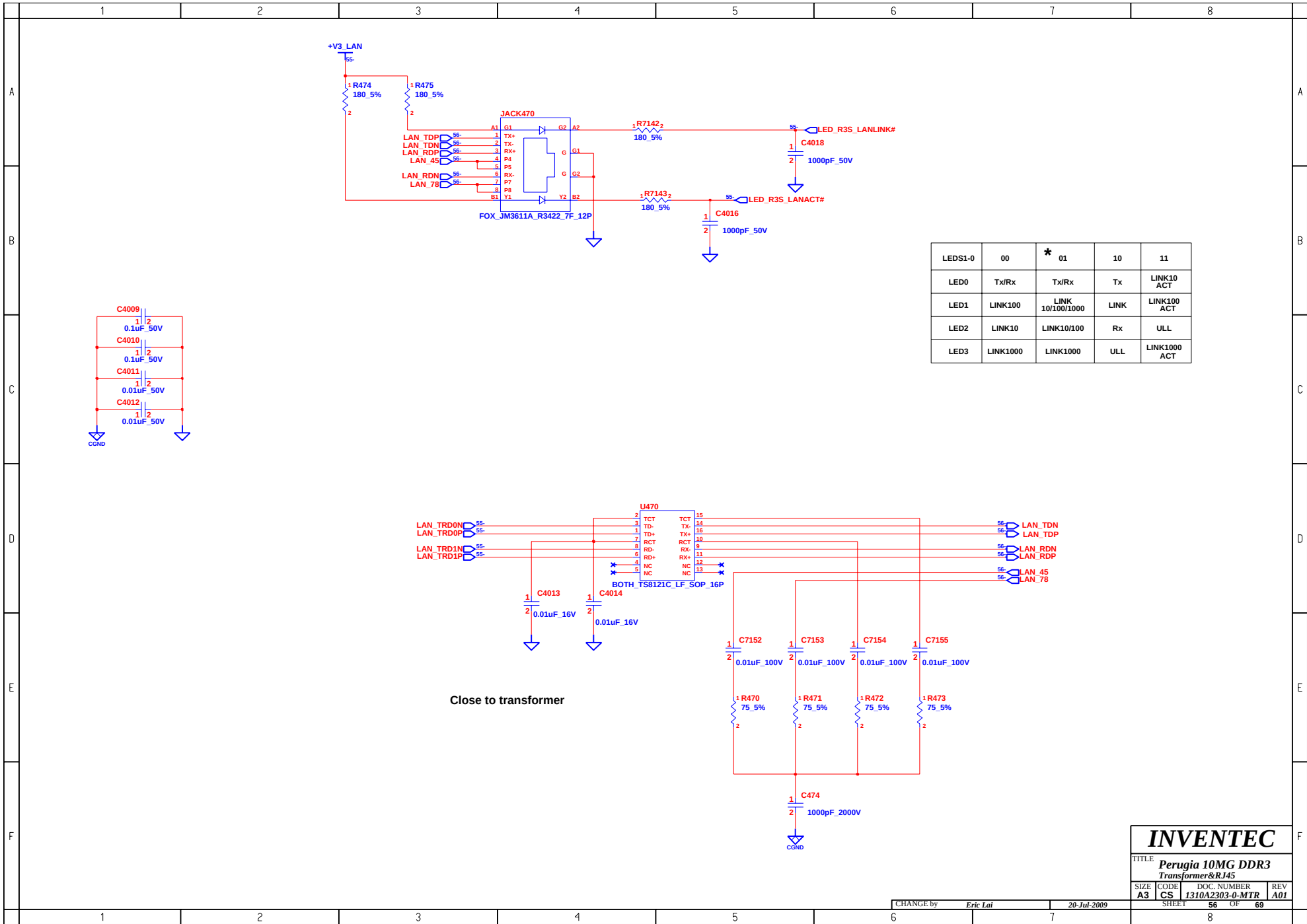
SIZE A3	CODE CS	DOC. NUMBER I310AZ303-0-MTR	REV A01
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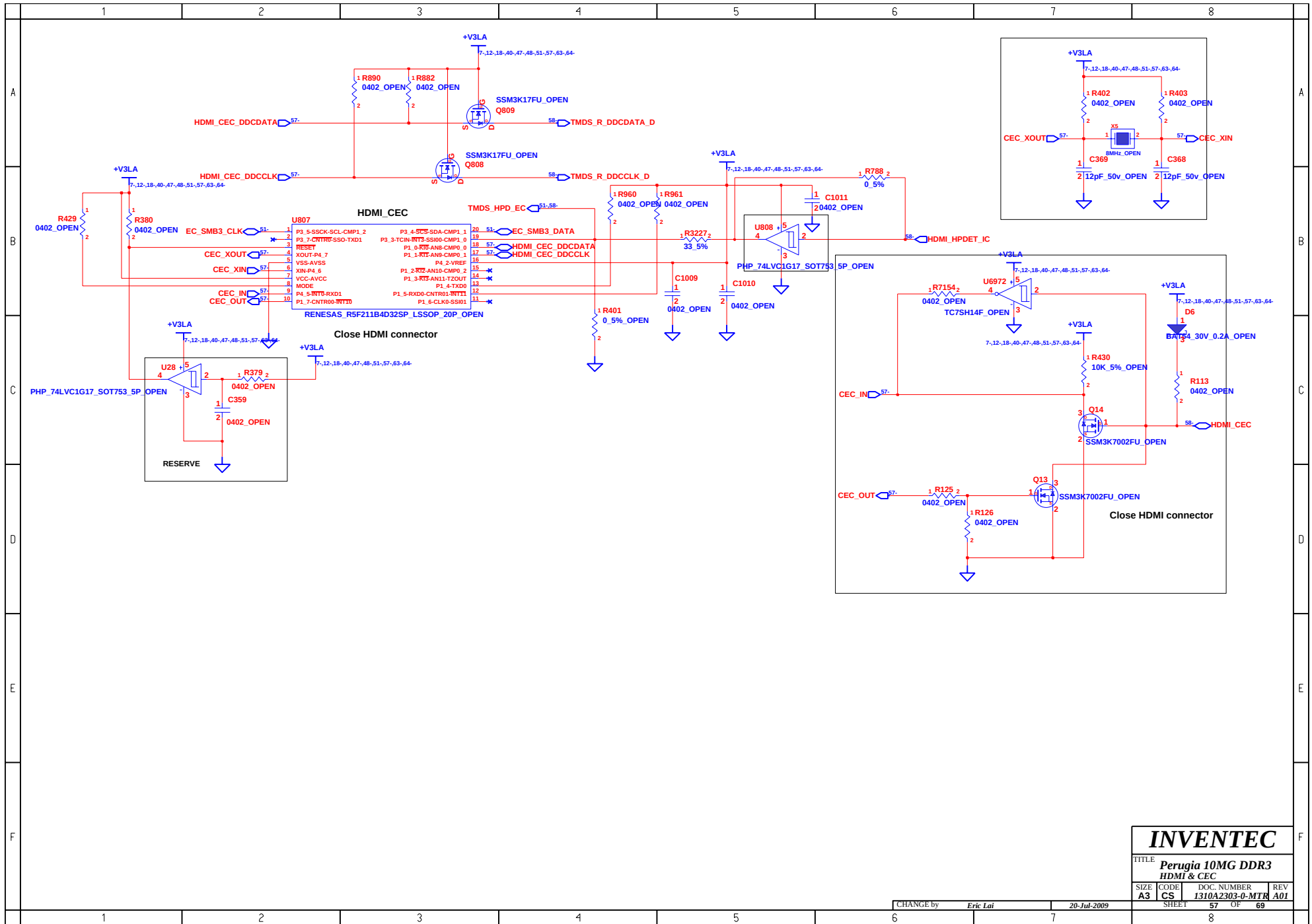
CHANGE by **Eric Lai** 20-Jul-2009

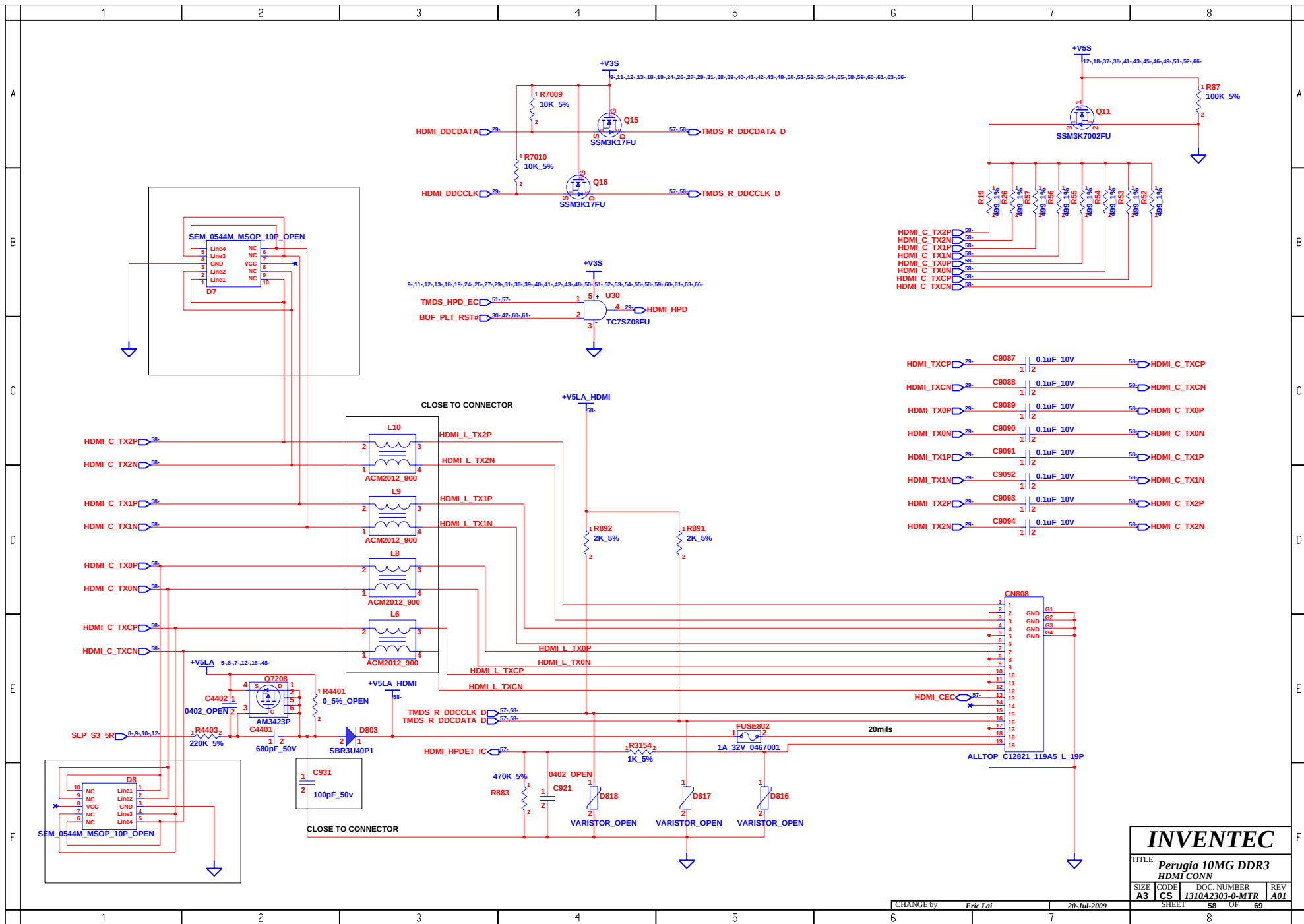
SHEET 52 OF 69











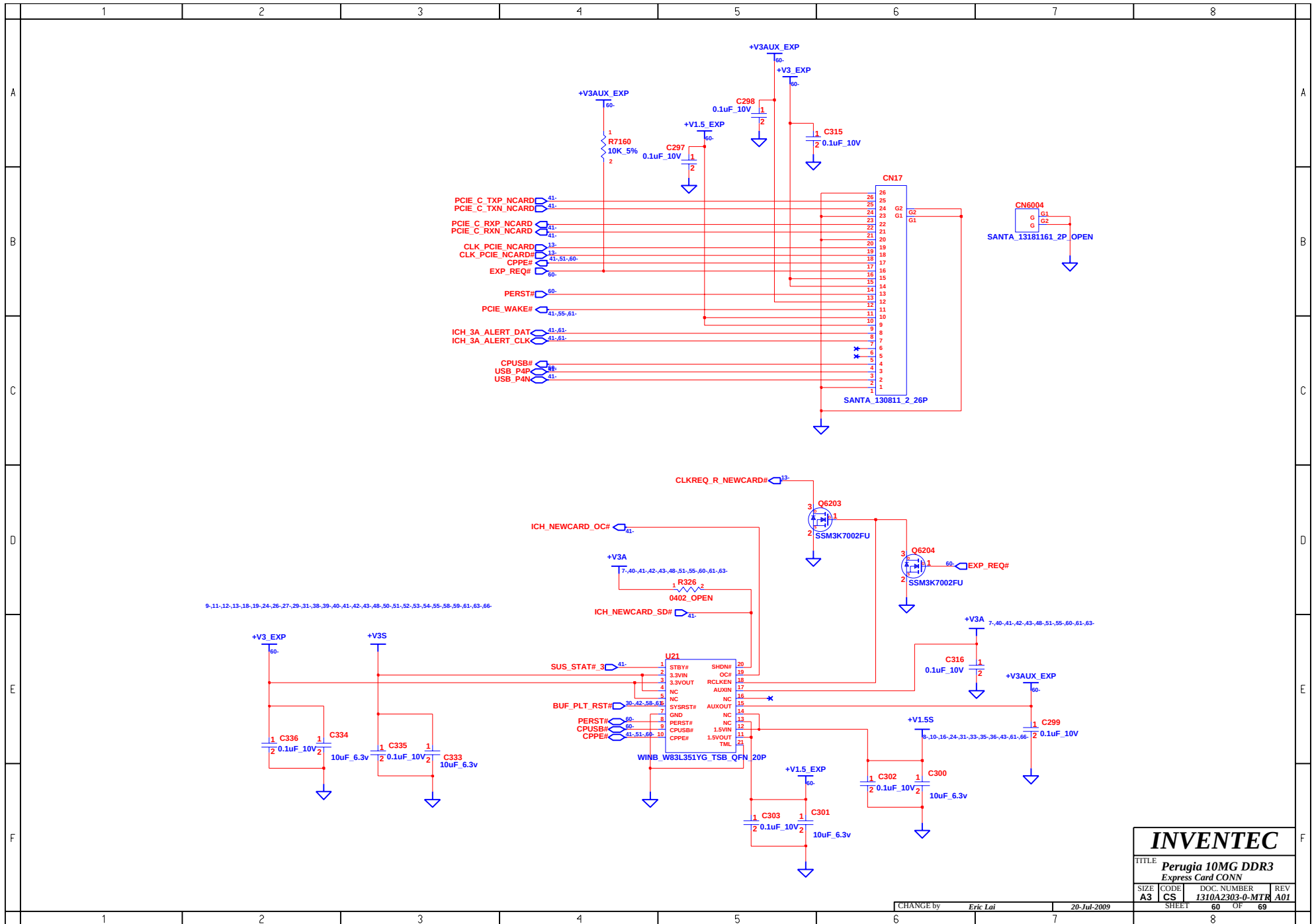
INVENTEC

TITLE
**Perugia 10MG DDR3
HDMI CONN**

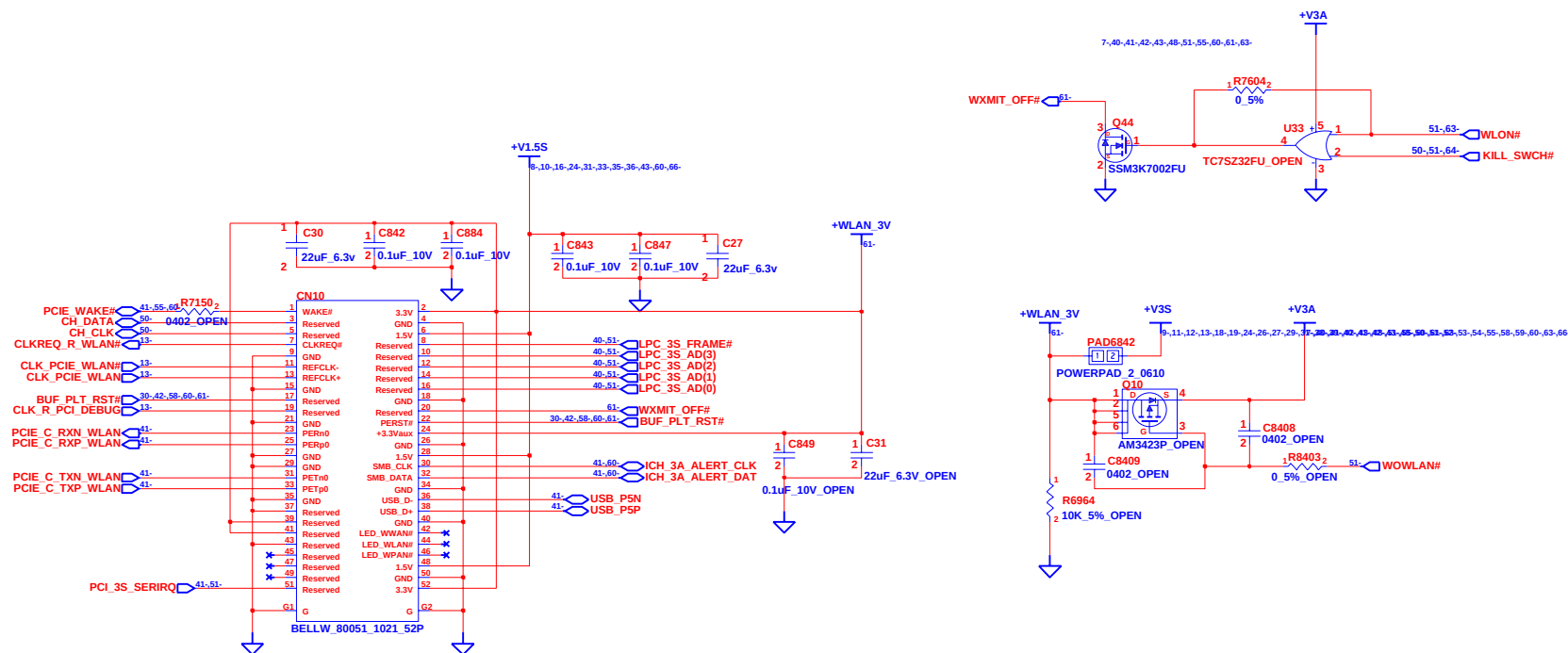
SIZE A3	CODE CS	DOC. NUMBER 1310A2303-0-MTR	REV A01
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CHANGE by *Eric Lai* 20-Jul-2009

SHEET 58 OF 69



Wireless & Debug card



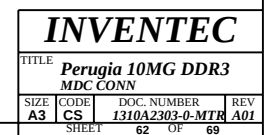
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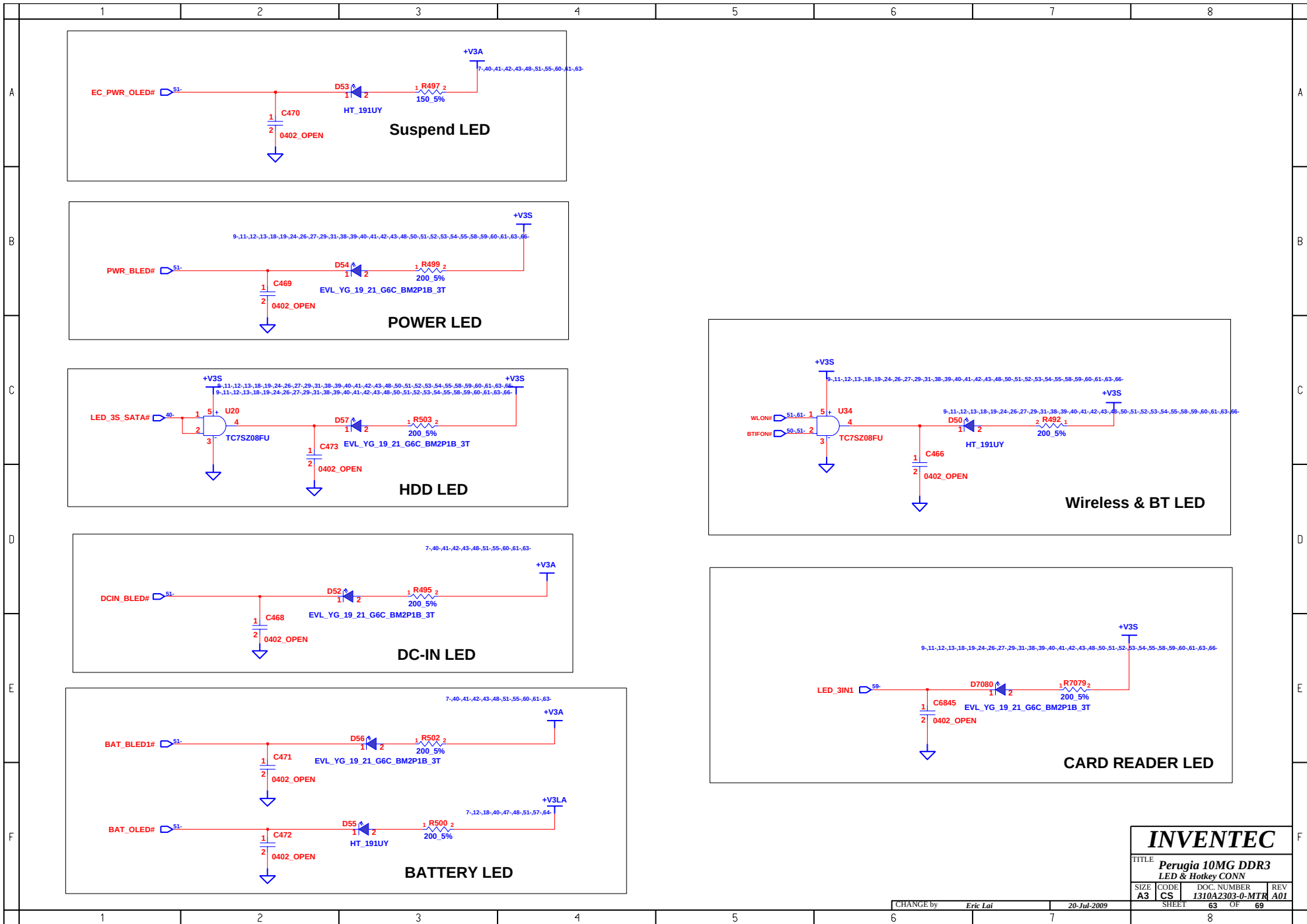
TITLE
Perugia 10MG DDR3
Wireless & debug card

SIZE	CODE	DOC. NUMBER	REV
A3	CS	I310AZ303-0-MTR	A01

CHANGE by Eric Lai 20-Jul-2009

SHEET 61 OF 69



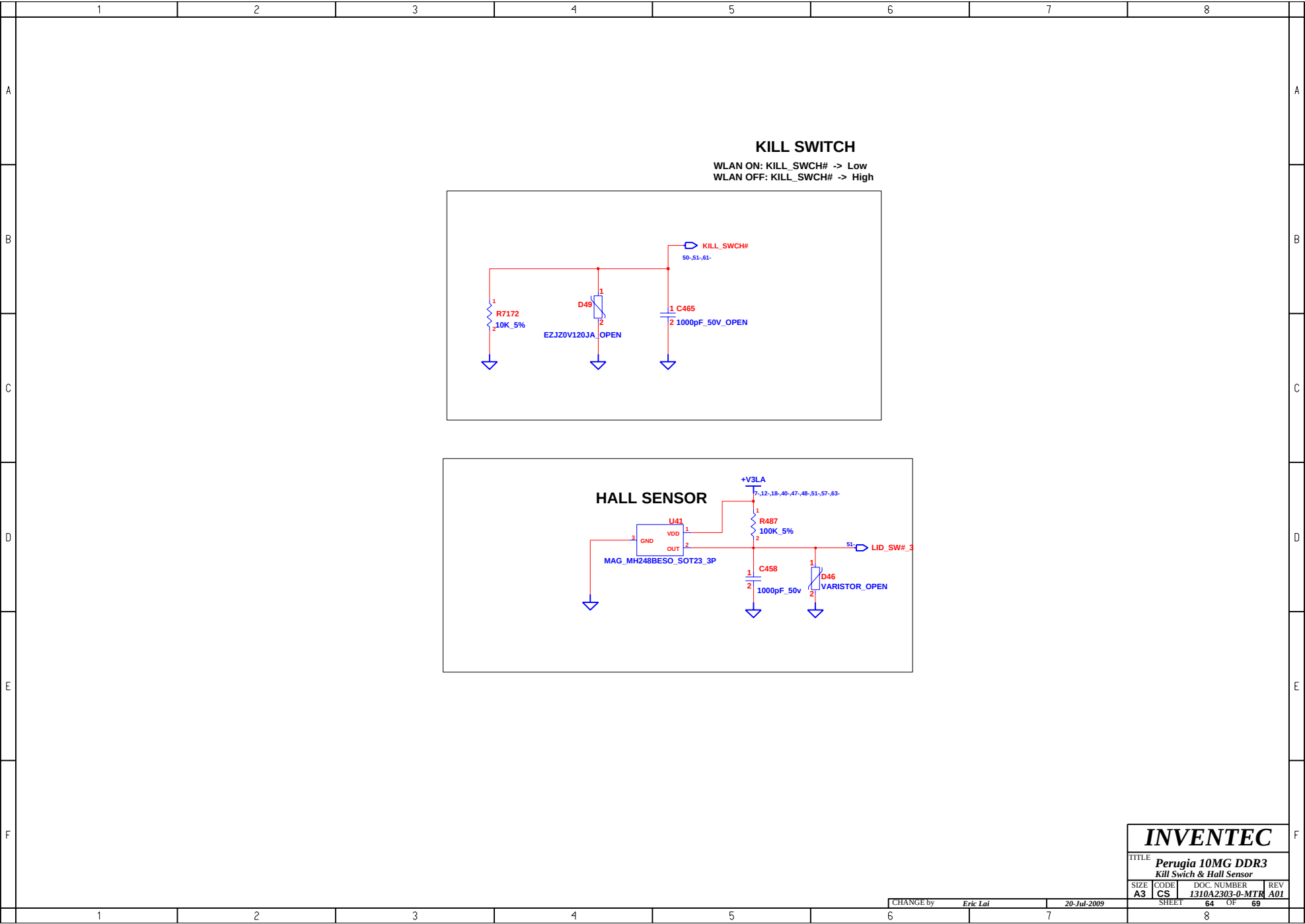


INVENTEC

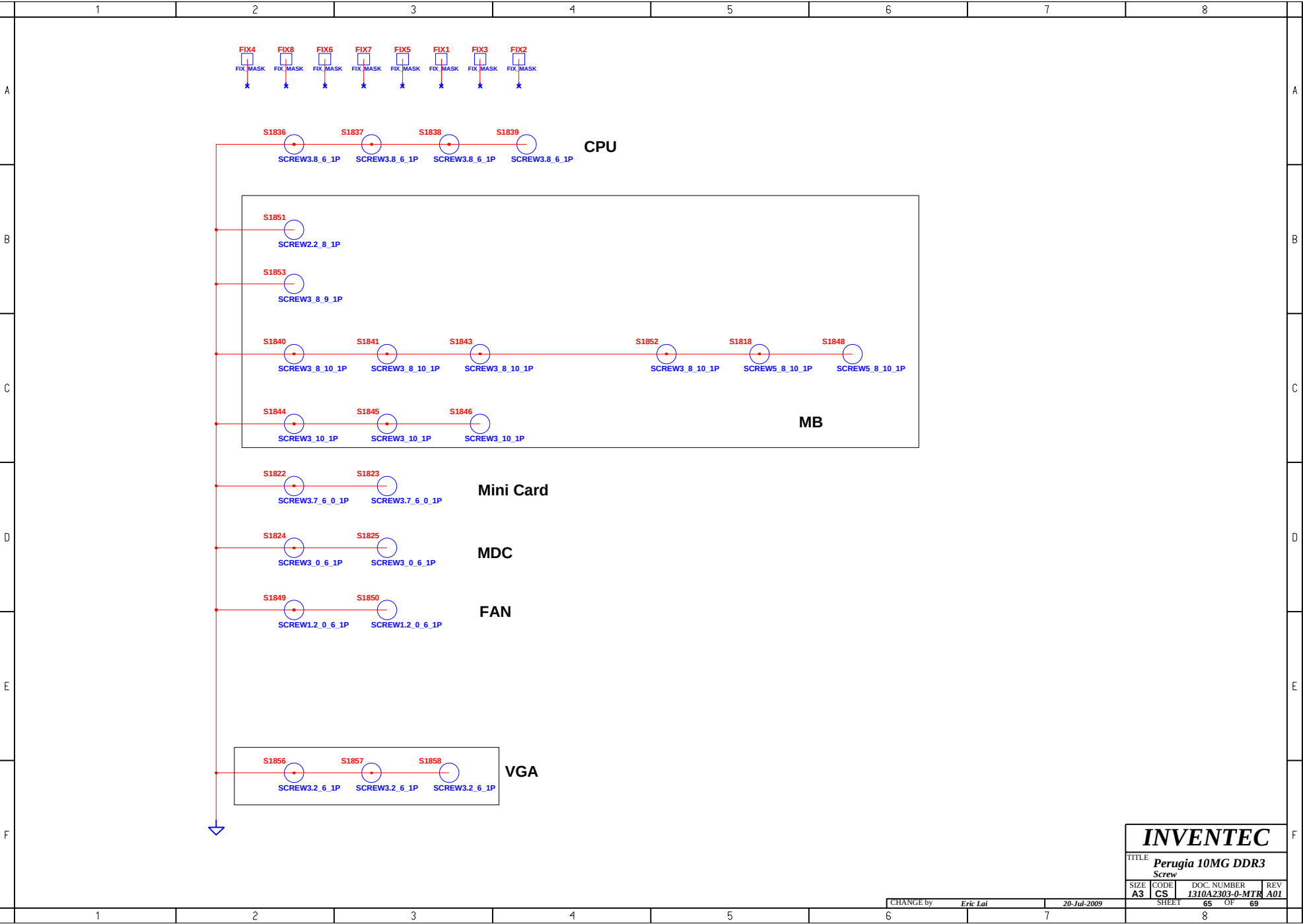
TITLE
**Perugia 10MG DDR3
LED & Hotkey CONN**

SIZE	CODE	DOC. NUMBER	REV
A3	CS	I310AZ303-0-MTR	A01

SHEET 63 OF 69



INVENTEC			
TITLE			
Perugia 10MG DDR3			
Kill Swich & Hall Sensor			
SIZE	CODE	DOC. NUMBER	REV
A3	CS	I310AZ303-0-MTR	A01
CHANGE by		Eric Lai	20-Jul-2009
SHEET		64	OF 69



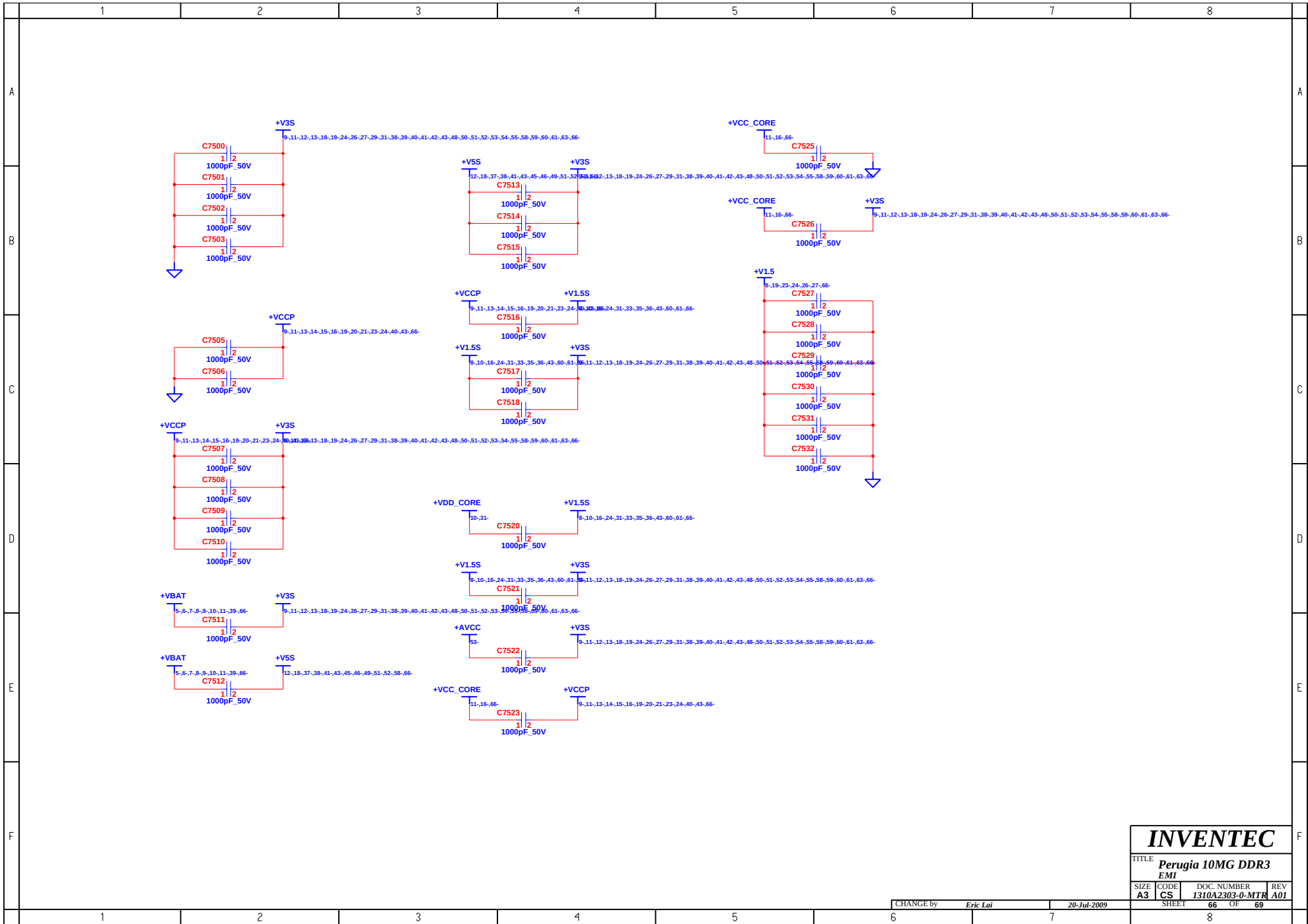
INVENTEC

TITLE
Perugia 10MG DDR3
Screw

SIZE A3	CODE CS	DOC. NUMBER I310AZ303-0-MTR	REV A01
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CHANGE by *Eric Lai* 20-Jul-2009

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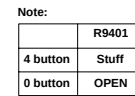
INVENTEC

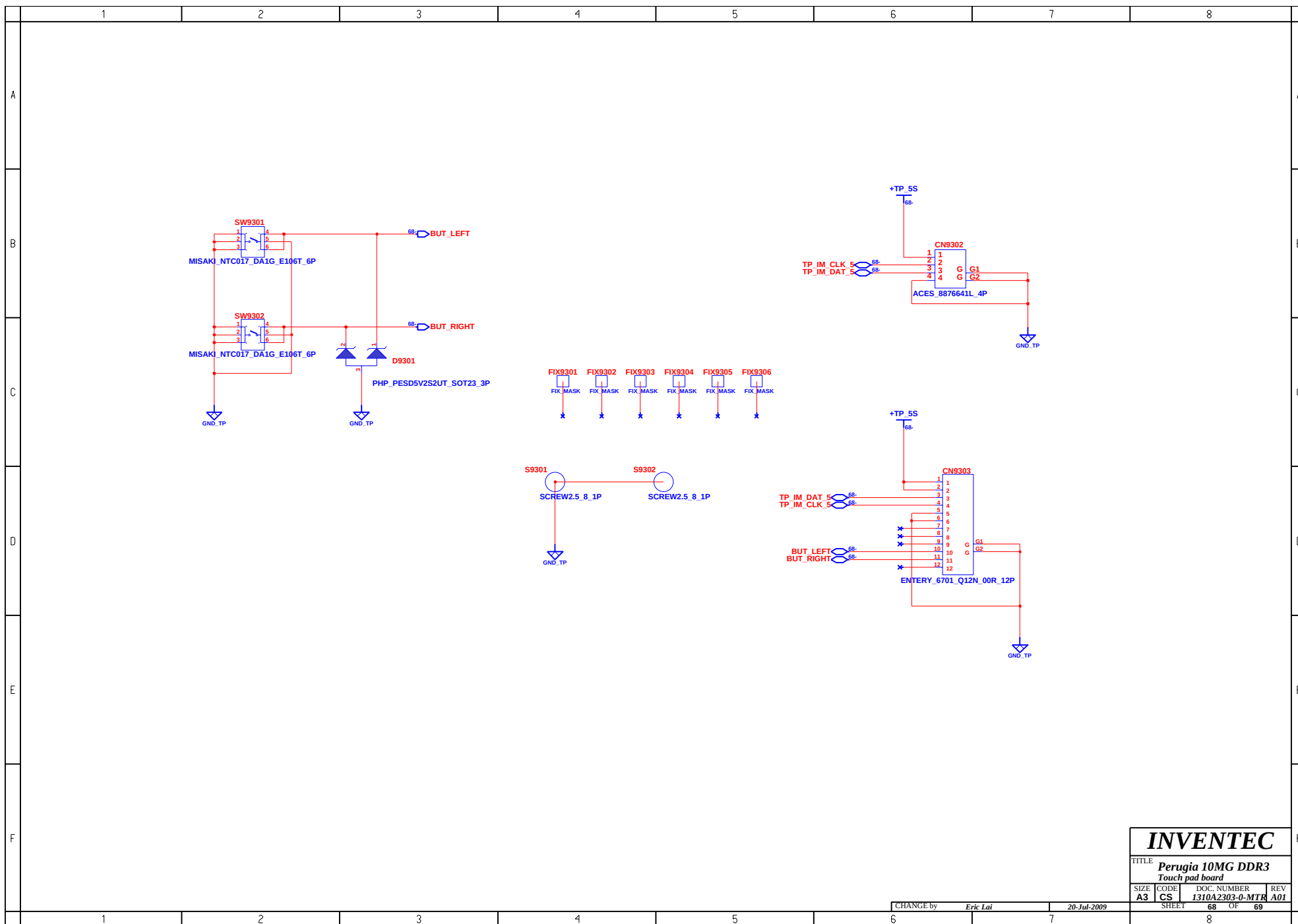
TITLE
Perugia 10MG DDR3
EMI

SIZE CODE A3 CS

DOC. NUMBER 1310AZ303-0-MTR

REV A01





INVENTEC			
TITLE			
Perugia 10MG DDR3			
Touch pad board			
SIZE	CODE	DOC. NUMBER	REV
A3	CS	I310AZ303-0-MTR	A01
CHANGE by		20-Jul-2009	
Eric Lai			
SHEET		68	OF 69

